

Bifolium Order in the Voynich Manuscript

A Diffusion Language Judge Side Quest

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Abstract

The Voynich Manuscript is bound as quires of nested sheets (bifolia). Lisa Fagin Davis has proposed, in a public lecture, that the sheets were meant to be read as *stacked* bifolia, rather than nested as they are currently bound. She also reported a collaborator's attempt to test this hypothesis with latent semantic analysis. The order of the text bears on the larger diffusion-guided decipherment project, but no published account of that test could be found, so an internal analysis had to be made.

This report describes an independent attempt to find order with a different instrument: the positions of *rare text*: the word types and glyph n -grams that occur 2–10 times, under a permutation null and known-text controls. It has three main findings.

- (i) The manuscript's rare text is about one tenth as local as that of prose and sits at the bottom of a corpus of known texts; this sets the power of every order test.
- (ii) Across the manuscript, the two leaves of one physical sheet share rare text more than leaves that are neighbours only in the binding. The result holds on words and on glyph n -grams, on both transcriptions, with quire, Currier language and scribal hand held fixed, and no other way of gathering a quire's sheets (up to 23,040 nesting patterns per quire) does as well as the best stacked order. **This suggests the interleaved (nested) binding is wrong: the manuscript is mis-bound at the sheet level**—or that the sheet was the unit of production; the text alone cannot tell the two apart (Section 10).
- (iii) The same statistic seriates the sheets inside a quire. Quire T settles on a single order of its six sheets whichever text unit or cost function is used. Quire C, in a different language and hand, likewise settles on one order; quire M is consistent with a single family of orders; the two small herbal quires are uninformative. Reading direction is not recovered. **This suggests the original order of the sheets within a quire can be recovered from the text, up to reversal.**

Findings (ii) and (iii) are unusually robust for quantitative work on the manuscript: the winning answers stand well clear of the field. The excess sharing between the two halves of a sheet is three to five standard deviations above what random placement of the rare text produces. This effect is seen in two different transcriptions and across multiple quires. In quire T the winning sheet order lies three to four standard deviations from the average of the 720 possible stacked orders and about four from the average of all 23,040 ways of gathering the sheets.

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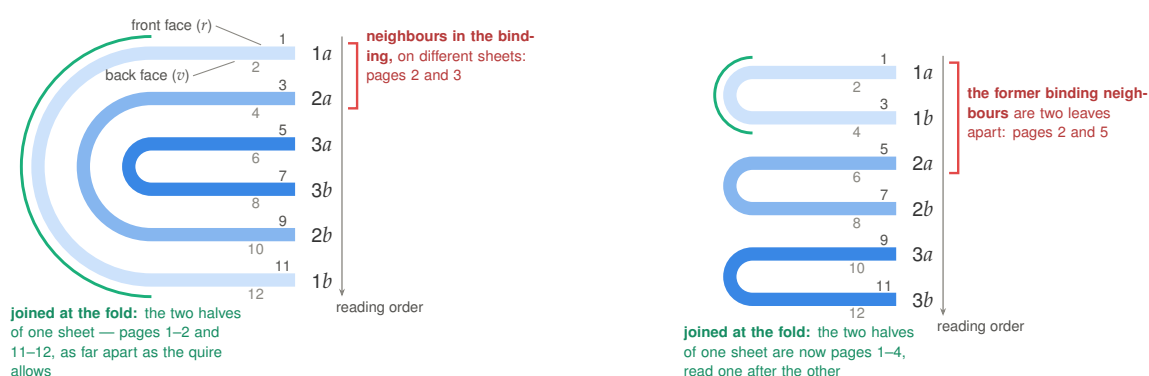
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Record status. Standalone write-up of docs/doubleton_gaps.md §2–19 as of 2026-09-03 (a side study of the diff-voyn project; no solver, judge or verdict of the decipherment programme depends on it). Every number below is traceable to a section of that record, marked [record: §N]; the record's later corrections (§16 on reversal-equivalence) are applied throughout. Data, figure code and this source live in docs/bifolia_report/. Throughout, *sheet* is used for what codicology calls a bifolium: one piece of parchment folded once into two leaves.

Executive summary

The question: stacked or nested?

The Voynich Manuscript is made of several folded vellum sheets known as bifolia. These bifolia are nested one inside another to form gatherings commonly referred to as quires. This construction results in the page order shown in Figure S1a, where the first folium of a bifolium is not adjacent to its second folium, except for the innermost bifolium. There has been some speculation that the manuscript was incorrectly bound some time after it was written. Lisa Fagin Davis has suggested, in a lecture at the Centre for Medieval Studies at the University of Toronto¹, that the Voynich Manuscript's bifolia were meant to be assembled one after another in *stacks* of folded bifolia as depicted in Figure S1b. She also reported that she was working with Colin Layfield to use Latent Semantic Analysis (LSA) to test the "out of order" hypothesis. Layfield and Davis recently released a paper exploring coherence and flow in the Voynich Manuscript using LSA², but a paper that addresses the codicological structure using LSA has not been published yet. Since understanding the intended order of the text in the Voynich is critical to the larger diffusion language judge project, we attempted to address the "out of order" hypothesis using a different technique that leverages rare text. This report describes the work done and the results that support Davis "out of order" hypothesis.



(a) **Nested:** three sheets nested one inside the next.

(b) **Stacked:** each sheet folded on its own, then the next.

Figure S1. A quire of three sheets seen from its top edge, spine at the left: nested as bound (a) and stacked (b). Each folded sheet gives leaves *a* (above the fold) and *b* (below); the small numbers are the reading order, down the pile, front face then back. The report's central test asks which pair of leaves shares more rare text: the two halves of one sheet, joined at the fold, or leaves that are neighbors only in the binding.

The instrument: where rare text recurs

In ordinary prose, a word that is used only two or three times typically clusters its uses over a few paragraphs or pages. If the pages were later assembled in a different order, these rare words would tend to move apart. As we describe below, the Voynich Manuscript has relatively few types of rare tokens that are found on multiple folia (see Figure S2 Top for an example). Although a signal is present, it is a little weaker than we would prefer. Luckily, the same effect is seen in rare strings of glyphs. These strings are between five to eight glyphs long, ignore

¹Link to video of lecture given September 26, 2025: <https://youtu.be/nH281tqYIyo?si=mhQL64gOIs475cNN&t=2122>

²Colin Layfield and Lisa Fagin Davis, "The Application of Latent Semantic Analysis to the Voynich Manuscript," *Digital Humanities Quarterly* 20, no. 1 (2026), <https://doi.org/10.63744/2ezxpskceq4>.

spaces, and can span words (see Figure S2 Bottom for an example). The use of rare glyph strings provides a stronger signal and does not assume that Voynich token is equivalent to a word.

We want a metric that describes how clustered these instances of rare text are. The measurement is the distance between two uses of the same rare word or glyph string along the reading order. For a whole text the ratio of Finding 1 is enough: short distances counted, divided by the count chance would give. To score a candidate order of the sheets, the plain average of the distances is poor, because it treats every rare word alike. A word whose uses cluster close together on one sheet, then appear once on another, says the two sheets were read together. A word spread over its sheet says little. So each stray use is scored as its distance to the word’s home sheet in units of that word’s home gap, the typical distance between its uses at home (Figure S3). Ten home gaps away costs ten, whatever the word. An order’s score is the sum over all the rare text, and the lowest wins.

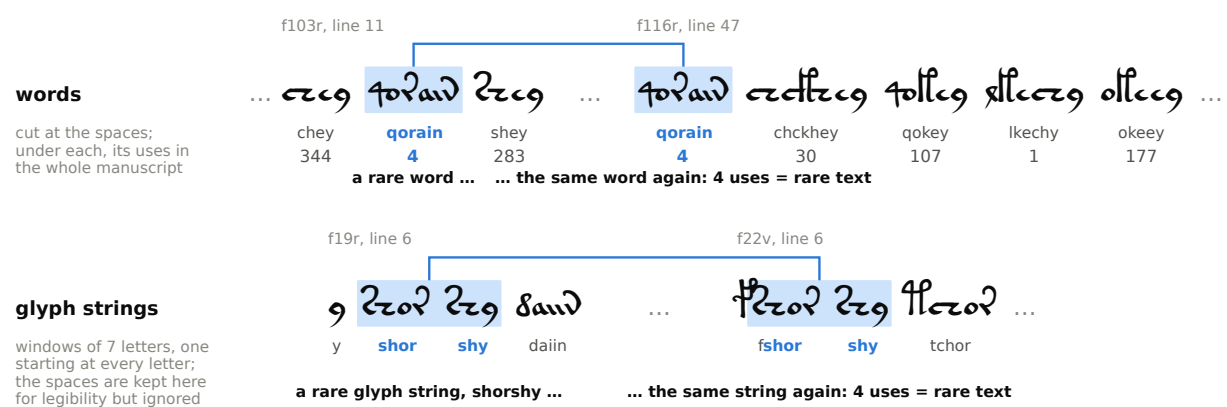
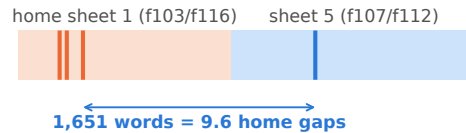


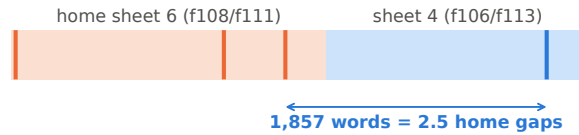
Figure S2. Two types of rare text are used in the recurring rare text distance measurements, Voynich “words” and Voynich glyph strings. Top, *words*: the Voynich word *qorain* occurs four times, and is separated by 9,500–11,900 words between successive uses (10,507 words between the two shown, on f103r and f116r). Bottom, *glyph strings*: the run of seven glyphs (with the spaces ignored) appears on f19r and 2,865 glyphs away on f22v inside a longer word.

qokaly, a clustering word

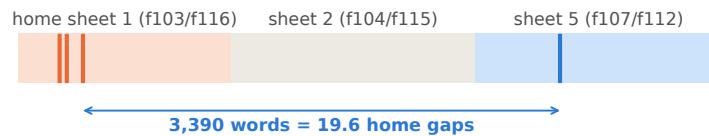
3 uses, 51 and 115 words apart
home gap 173 words

**chckhedy, a spread word**

3 uses, 1,480 and 437 words apart
home gap 756 words

**qokaly again, another order**

sheet 2 read between
same home gap, 173 words



use on the home sheet

stray use, on another sheet

score of a candidate order = each stray use's distance in home gaps, summed over the quire's rare text; the lowest score wins

Figure S3. The cluster-scaled score on two words of quire T. Each row is read left to right as one candidate order: the word's home sheet, then the sheet holding its one stray use. Ticks are uses. The spacing of the uses at home sets the word's home gap, and the stray costs its distance to the nearest use at home divided by that gap. *qokaly* clusters and *chckhedy* spreads, so their strays sit at almost the same distance but cost 9.6 and 2.5 home gaps. The plain average would count them alike. The third row reads sheet 2 between the two: the distance and the cost grow, the home gap does not, because it is set on the home sheet alone. An order's score sums the cost of every stray use of rare text in the quire.

Finding 1 — The manuscript's rare text hardly clusters at all

In known prose, a twice-used word's two occurrences fall within 100 words of each other 13 to 51 times more often than chance; in the manuscript, under four times (Figure S4, which pools words used two to five times). Against 148 windows of known text cut to the manuscript's length, the manuscript sits at the extreme low edge, with only Tasso and Dante below it; the pattern is the same for words used three, four or five times, and in both of the manuscript's "languages" (Currier A and B). Nor is this an artefact of the binding: re-ordering the manuscript's 52 sheets to make its rare text cluster as much as any order can lifts the figure only from 3.9 to 4.5, and a manuscript whose page contents were shuffled first is lifted by the same amount (Section 5.3). Its rare text clusters like the rare rhyme-words of narrative verse, not like the topic words of prose.

What this means. Two things. First, every test of order in this report has to find its signal in a text with about a tenth of the locality of the control texts, so the tests are weak, and where a result is null that is the expected outcome rather than evidence of absence. Second, this is a property of the text itself, whatever it is: its rare "words" do not behave as topic words do.

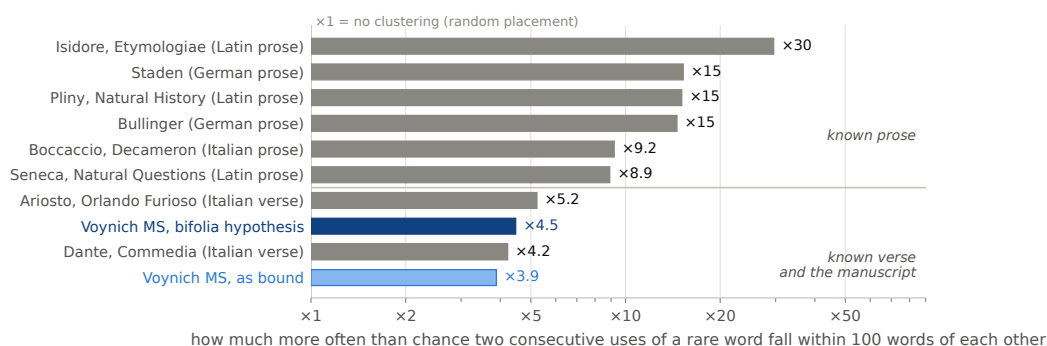


Figure S4. Rare-text clustering, relative to chance: for words used two to five times, how often two consecutive uses fall within 100 words of each other, divided by the same rate under random placement. Known texts are windows of the manuscript’s length. Prose clusters 9–30 times more than chance, verse 4–5, the manuscript as bound 3.9 (4.4 in the other transcription). The *bifolia hypothesis* bar is the best of every re-ordering of its 52 sheets, each read whole: 4.5, the same gain the optimiser finds when the page contents are shuffled first. (Section 3, Section 5.3; Table 1, Table 6.)

Finding 2 — The two halves of a sheet belong together

Despite the weak signal, the two halves of a physical sheet share rare text more than the pages that are neighbors only in the binding do—and the binding-neighbours share *less* than random pages of the same quire (Figure 4a in the main text). On rare word types the excess is 2.4–2.5 standard deviations above chance; on rare seven-glyph strings, an independent unit run as the pre-stated check, it is 4 to 5. The six control texts settle what that size means: written sheet by sheet they land in the same band as the manuscript, written in the binding order they land far on the other side, on all 36 of 36 comparisons each way. Two objections were tested. The scribes’ hands run by sheet (97 % of sheet halves are in one hand), so a scribe’s private spelling habits could mimic the effect: holding the hand fixed in the comparison leaves the glyph-string result unchanged. And an excess this large might arise by accident: in 5,000 random re-placements of the pages it never did, on either transcription. One caution belongs here: the test on words used exactly twice was null, the pooled word statistic was chosen after that, and the glyph-string test was then run as the check; the glyph-string number is the one to quote.

What this means. The text on one sheet is a unit. What this does *not* decide is whether the sheet was the unit of *reading* (Davis’s proposal) or the unit of *work*—one scribe, one sitting, one subject per sheet, with the nested reading order still intended. Every statistic in this report responds identically to the two.

Finding 3 — No other way of gathering the sheets fits as well

Finding 2 compared the sheets read whole with one nested arrangement, the binding. Perhaps a different nesting, or two smaller gatherings placed one after another, would explain the text as well. For the six-sheet quire T every such pattern was scored, 23,040 in all. The fully nested orders are the worst group and a tight one, no member of it fitting better than an average pattern, while the stacked orders are the best group and the best of them is the best of all 23,040 (Figure S5); the mixed patterns lie in between, the fit improving at every step from “one nested block” toward “every sheet read whole” (Figure 15 in the main text). On glyph strings the best fully nested arrangement ranks between 5,007th and 10,535th, the binding as it is between 17,902nd and 22,405th (Table 8 in the main text), and no pattern with two or three nested groups ever wins. The same holds, with smaller candidate sets and coarser odds, in the other four quires tested.

What this means. The property the text rewards is exactly the one every stacked order has,

a sheet's four pages kept together, and the comparison in Finding 2 was not a straw man: the binding is not merely worse than the best alternative, it is among the worst of all arrangements.

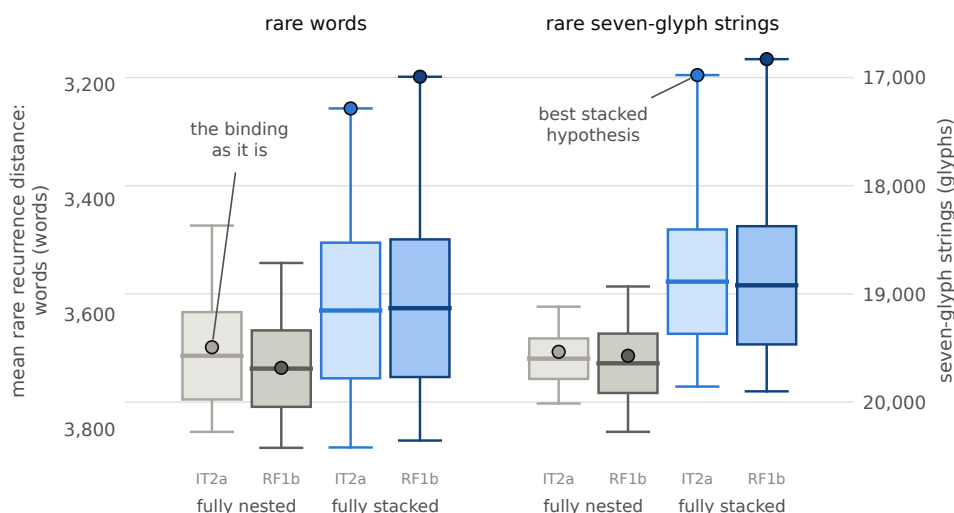


Figure S5. The binding compared in Finding 2 was no straw man: all 720 ways of nesting quire T's six sheets (grey; the binding dotted) fit the rare text as poorly as it does, while the 720 orders reading each sheet whole (blue) fit better on average and their best (dotted) is the best of all 23,040 gatherings on glyph strings, 5th or 2nd on words (closer fit higher). The wide spread among stacked orders is what Finding 4 exploits. Boxes mean $\pm$ 1 sd, whiskers extremes, shades IT2a/RF1b. (Section 7.)

Finding 4 — Inside quire T the sheets have a preferred order

If a sheet is a unit, the sharing between sheets can say which sheet followed which. For the six sheets of quire T there are 720 orders; one of them, 1-6-5-4-2-3 (the outermost sheet beside the innermost, the rest running outward), is chosen by 31 of 32 combinations of text unit, cost function and transcription (Figure 12a in the main text; the orders and folios of all five quires are listed in Table S1). It fits four standard deviations better than the average order, where the best of 720 orders on shuffled page contents sits near three; against those shuffles its odds are at most 1 in 50 on every test, and the size of the sheets alone cannot produce it. The current bound order lies in the bottom half. Quire C, a herbal quire in the other Currier language and another hand, settles the same way on 1-4-3-2 on 30 of 32 tests; quire M is consistent with one family of orders; the two smallest herbal quires (1,000–1,500 words) give no single order. In each of T, C and M the outermost sheet neighbours the innermost.

What this means. Within at least one quire the text remembers an order of its sheets, and it is not the bound order. What it does not give is the *direction* of reading along the chain: the gap between the winning order and its reversal is no larger than selection alone produces on shuffled pages, and there is no time arrow inside a cluster of rare text to calibrate on (prose clusters are symmetric to within a per cent). Nor can the order be attributed to reading rather than to working sheet by sheet, for the reason given under Finding 2.

quire	sheets as bound (outer → inner)	order the text prefers	evidence
T — stars, f103–f116 Currier <i>B</i> , hand 3 10,700 words			31 of 32 tests; odds ≤ 1 in 50 vs shuffled pages on every test; bound order ranks 150th–626th of 720
C — herbal, f17–f24 Currier <i>A</i> , hand 1 1,400 words			30 of 32 tests; only 24 candidates, so the odds are coarse (≤ 1 in 20 on most tests); bound order ranks 4th–10th of 24
M — biological, f75–f84 Currier <i>B</i> , hand 2 6,900 words			one family of orders on glyph strings (also 3–2–5–1–4 and 4–1–2–3–5; odds ≤ 1 in 20 on 14 of 16 tests), not on words; bound order ranks 30th–114th of 120
A — herbal, f1–f8 Currier <i>A</i> , hand 1 1,500 words		stacked, but no single order: the winner changes with the text unit	sheets read whole beat the binding (best stacked order 1st–3rd of 192 gatherings, the binding 31st–183rd), but not beyond what shuffled pages give (P 0.09–0.36); bound order ranks 10th–20th of 24
B — herbal, f9–f16 Currier <i>A</i> , hand 1 1,000 words		stacked on glyph strings; no single order	on glyph strings the best stacked order ranks 1st of 144 gatherings on all eight tests, the binding 46th–125th (on words the binding wins); no order beats shuffled pages: too little text (f12 is lost)

Table S1. Sheet orders by quire. Sheets are numbered from the outside inward as bound (1 = outermost) and coloured by that position, as in Figure 11; the folios beneath are the sheet’s two leaves. Each sheet is read whole, front leaf then back, and the chains give *adjacency only*: either end may be the beginning (Section 8).[†]

[†] In quire T the winner puts f103/f116 at the *end* of the chain on 22 of 24 tests, but the gap to the reverse reading is what selection alone gives on shuffled pages (p 0.2–0.8), so no direction is claimed.

What the report establishes, and what it does not

What this establishes

- **The sheet is a unit of the text.** The two halves of a physical sheet share rare text more than pages that are neighbours only in the binding, on words and on glyph strings, in both transcriptions, with quire, language and scribal hand held fixed; real texts reproduce this only when written sheet by sheet (Finding 2).
- **The binding as it is fits the text worse than almost any other arrangement of the same sheets,** and nothing but reading the sheets whole fits as well (Finding 3).
- **Quire T has a recoverable sheet order,** 1–6–5–4–2–3, replicated in shape by quire C (Finding 4, Table S1).
- **The manuscript’s rare text carries about a tenth of the locality of prose** (Finding 1), which is why the effects above are small in absolute terms: three quarters of twice-used words span more than one quire, and no re-ordering of the manuscript’s 52 sheets makes its rare text cluster like that of prose.

What it does not establish

- **That the manuscript was meant to be read sheet by sheet.** “Written sheet by sheet as a unit of work, intended to be read nested” produces every observation above. Text position cannot separate the two; production evidence (ink, ruling, the sequence of drawings) could.
- **The direction of reading** along any chain of sheets.
- **Which sheets or quires of the whole manuscript are out of place.** Five quires were seriated; the mixed-language quires, the large foldouts and the codicological literature’s specific proposals were not tested.
- **Anything about the language or a decipherment.** This is a side study; no solver or verdict of the main project depends on it, and the stacked order does not change the solver’s inputs.

Table S2. Reading guide: the claims, how strong the evidence is, and where to find it.

claim	strength	basis	where
rare text clusters at $\approx 1/10$ of prose	●●●	148 known windows of equal length; both transcriptions	Section 3
sheet halves share more than binding neighbours	●●●	+3 to +5 sd on glyph strings; 0/5,000 random placements; 36/36 controls each way; hand-controlled	Section 4
the effect is small in mass; no global re-ordering helps	●●●	rare text not contained by sheet; optimiser gains only its noise ceiling	Section 5
every other gathering of the sheets loses to reading them whole	●●● / ●●●	decisive in quire T (23,040 patterns); coarser in M and the four-sheet quires	Section 7
quire T order 1-6-5-4-2-3	●●●	31/32 tests; odds ≤ 1 in 50 vs shuffled pages on every test; geometry excluded	Section 6
quire C order 1-4-3-2	●●●	30/32 tests, but only 24 candidates, so the odds are coarse	Section 6
quire M order family	●●●	consistent across units; words weak	Section 6
quires A and B	●●●	too little text; no information	Section 6
reading direction along a chain	●●●	gap to the reversal is what selection alone gives; no time arrow in clusters	Section 8
reading unit vs. working unit	●●●	not testable from text position	Section 10

●●● strong ●● moderate ● weak ●●● null / not established ●●● outside what the method can test

1 Question and codicological setting

A *sheet*—a bifolium, in the codicologist’s term—is folded once into two *leaves* a and b , giving four *pages* a^r, a^v, b^r, b^v . A quire of S sheets nests them: sheet 1 is outermost, sheet S innermost. Read as bound, the page sequence is

$$1a^r, 1a^v, 2a^r, 2a^v, \dots, Sa^r, Sa^v, Sb^r, Sb^v, \dots, 2b^r, 2b^v, 1b^r, 1b^v, \quad (1)$$

so the two leaves of an outer sheet are far apart and consecutive leaves mostly belong to different sheets. Read *stacked*, each sheet is read whole,

$$1a^r, 1a^v, 1b^r, 1b^v, 2a^r, 2a^v, 2b^r, 2b^v, \dots \quad (2)$$

and the two leaves of a sheet are always adjacent (Figure 1). Two kinds of leaf pair therefore discriminate the arrangements: *conjugate* leaves (the two leaves of one sheet, adjacent only under stacked reading) and *nested-adjacent* leaves (consecutive in the binding but on different sheets, adjacent only under nested reading).

If a text was written continuously, words and phrases that belong to one passage sit close together *in the writing order*. The recurrences of rare text are the cleanest carrier of that locality: a topic word used twice in 38,000 words is used twice in the same passage. So the question “was the text written in the nested order or the stacked order?” becomes “do conjugate leaves or nested-adjacent leaves share more rare text than chance?”—a question with a permutation null and known-text controls. Two further questions follow once the first is answered: whether the sheet sharing can *order* the sheets of a quire (seriation), and whether it can give the *direction* of reading along that order.

Provenance of the proposal, and a prior attempt. The stacked-sheet proposal is, to our knowledge, not in print. Davis put it forward once, in a public lecture,³ and said there that a colleague at the University of Malta was trying to test it against the text with latent semantic analysis (LSA)—that is, whether the topical similarity of pages favours the stacked over the nested sequence. We have found no journal or preprint account of that analysis, so its statistic, its controls and its outcome are unknown to us, and nothing below builds on it. The present study is therefore not the first attempt to look for the arrangement in the text but a second one, made independently and with a different instrument: not topic vectors over whole pages but the positions of individual rare types, a permutation null that holds quire, Currier language and scribal hand fixed, and six known texts written onto the manuscript’s own page slots as positive and negative controls. If the LSA result becomes available, the point at which the two approaches should agree is the sign of the conjugate-versus-nested contrast of Section 4; they need not agree on the within-quire seriation of Section 6, which uses a finer unit than any page-level topic model.

2 Data, units and notation

Streams. Both public EVA transcriptions are used throughout: Takahashi (IT2a) and the Reference transcription (RF1b). Tokens containing an uncertain glyph are dropped whole; extended-EVA codes and ligature braces are kept as part of the word type. In folio order the streams hold 37,759 (IT2a) / 38,463 (RF1b) tokens on 225 / 227 pages, mean 168 tokens per page; 896 / 997 doubleton types and 5,527 / 6,685 hapax types. Page headers supply quire, sheet (bifolium), Currier language (A/B), section and scribal hand (Davis’s five scribes). [record: §1]

³Recorded lecture, online video (c. 2023–24); the link will be added to this report once retrieved. [URL pending.]

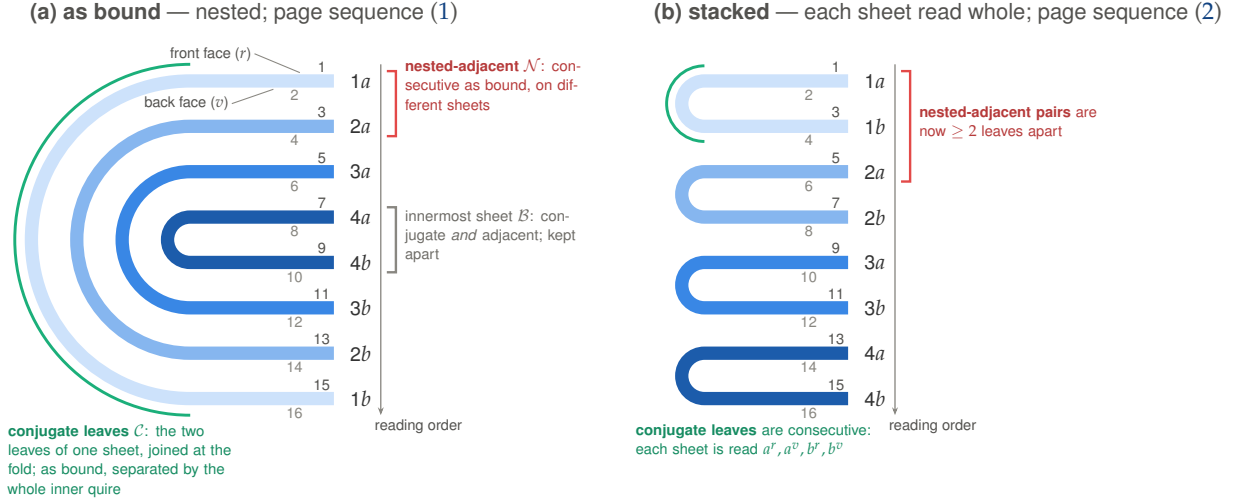


Figure 1. Nested versus stacked reading of a four-sheet quire, seen from its top edge with the spine at the left. Leaf a of each sheet lies above its fold and leaf b below; the numbers give the reading position of each leaf’s front (r) and back (v) face, read down the pile. (a) The binding, Equation (1): conjugate leaves C —the two leaves of one sheet, joined at the fold (aqua)—are separated by the whole inner quire, while the leaves that follow one another lie on different sheets (nested-adjacent pairs $\mathcal{N}$, red); the innermost sheet’s two leaves are both at once ($\mathcal{B}$) and are excluded from the statistic. (b) The stacked order, Equation (2): each sheet is read a^r, a^v, b^r, b^v , so conjugate leaves are consecutive and the former nested-adjacent pairs are at least two leaves apart. A text written in one order and bound in the other leaves its rare-text locality on the “wrong” leaf pairs.

Rare types. A *type* is a word type or, for the sub-word analyses, a character n -gram ($n = 4\text{--}8$) over the space-stripped page text, with extended-EVA codes as single symbols. A type t with occurrence set O_t is *rare* when $k_t = |O_t| \in [2, 10]$; $k_t = 2$ is a doubleton. The occurrences of all rare types of a stream, whatever the unit, are its *rare text*, the material every test below counts. The variant *cross* keeps only n -grams that straddle a word boundary, so that they are fragments of word *pairs* rather than of single rare words. Figure 2 shows both units on one line of the manuscript and where that line’s rare text recurs. [record: §7]

Known texts. Latin (Isidore, Seneca *NQ*, Pliny, Macer, Gordon, Hyginus, Bacon), German (Bullinger, Staden, Promptuarium, Küchenmeisterei) and Italian (Decameron, Principe, Commedia, Orlando Furioso) texts from the project corpus, cut into windows of 37,759 tokens and pseudo-pages of 168 tokens; for the corpus sweep, every corpus document of $\geq 37,759$ words (148 windows: German 113, Latin 25, Italian 10). [record: §1, 9]

Notation. Pages are slots $p = 1..P$ with contents c_p (a bag of typed occurrences) and token length L_p . $q(p)$, $\lambda(p)$, $h(p)$, $s(p)$ and $\ell(p)$ denote quire, Currier language, hand, sheet and leaf. An *occurrence pair* is (i, j) , $i < j$, two occurrences of the same rare type on different pages. For a page order σ , $\text{pos}_\sigma(i)$ is the token index of occurrence i and $d_\sigma(i, j) = |\text{pos}_\sigma(i) - \text{pos}_\sigma(j)|$.

Empirical p -values. Every p quoted below is a Monte-Carlo tail with the add-one convention,

$$p = \frac{\#\{\text{null draws at least as extreme as observed}\} + 1}{n_{\text{draws}} + 1}, \quad (3)$$

so with 200 draws the floor is $1/201 \approx 0.005$ and with 5,000 draws $\approx 2 \times 10^{-4}$. A z is $(\text{obs} - \bar{x}_{\text{null}})/s_{\text{null}}$; where the null’s shape matters it was checked directly (Section 9).

(a) one line of the manuscript, cut into words and into glyph strings



(b) where this line's rare text recurs: on the other half of the same sheet



Each such shared occurrence is one conjugate-leaf pair in the leaf-pair test. The same strings recur on other pages too; the test asks only whether pairs of occurrences fall on the two halves of one sheet more often than random placement gives.

Figure 2. Rare text on one line of the manuscript (f116r, line 47, Takahashi transcription), in the manuscript's glyphs with the EVA transliteration beneath. (a) Cut into words, the line has nine word types; the number under each is k_t , its count in the whole manuscript, and *osain* and *qorain* are rare ($2 \leq k_t \leq 10$). Cut into glyph strings, the spaces are removed and a 7-letter window starts at every letter (extended-EVA codes and each letter of a bench such as *ch* are single symbols); the strip classifies all 44 windows of the line by k_t , and five are called out. Most windows straddle a former word gap—these are the *cross* variant—so a rare glyph string is usually a fragment of a word pair and needs no theory of where a word begins or ends. (b) Three of the line's rare glyph strings, and its rare word *qorain*, occur again on f103r, the conjugate leaf of the same sheet (sheet 1 of quire T), joined to f116r at the fold and 25 pages away in the binding. Each such shared occurrence is one $\mathcal{C}$ pair in the leaf-pair test of Section 4; the same strings recur on other pages too, and the test asks only whether occurrence pairs fall on conjugate leaves more often than the permutation null gives.

3 How local is the manuscript's rare text?

The power of every order test is set by how tightly a text's rare types cluster in its writing order. For a type with k occurrences at positions $x_1 < \dots < x_k$ in a stream of N tokens, the consecutive gaps $g_m = x_{m+1} - x_m$ are compared with their distribution under uniform placement of k points on N (exact for $k = 2$; 200,000 Monte-Carlo draws for $k \geq 3$). The locality ratio at

Table 1. Rare-text locality r_{100} by frequency class k (in brackets: percentage of types with all occurrences on one page). Manuscript rows are the whole stream; Currier *A* and *B* alone are 11 k and 23–24 k tokens. [record: §2, 6.1]

text	$k = 2$	$k = 3$	$k = 4$	$k = 5$
VMS IT2a (types 896 / 399 / 216 / 145)	3.8 (2.9%)	3.3 (0.3%)	3.5 (0%)	4.5 (0%)
VMS RF1b (997 / 413 / 250 / 157)	3.9 (2.3%)	3.7 (0.2%)	5.0 (0.8%)	4.4 (0.6%)
Currier <i>A</i> alone (IT2a / RF1b)	2.4 / 3.1	2.7 / 2.9	2.9 / 3.2	3.2 / 2.6
Currier <i>B</i> alone	2.6 / 3.2	3.3 / 3.0	2.9 / 3.0	3.0 / 1.9
Isidore (Latin prose)	51 (23%)	32 (7.0%)	23 (2.2%)	18 (0.5%)
Pliny (Latin prose)	21 (10%)	19 (3.5%)	12 (0.3%)	12 (1.3%)
Seneca <i>NQ</i> (Latin prose)	13 (5.4%)	9.5 (1.9%)	8.6 (0.3%)	6.0 (0%)
Bullinger, Staden (German prose)	23–26 (11%)	16–17 (3%)	14–15 (0.5–1.3%)	9–11 (0.6–0.8%)
Decameron (Italian prose)	13 (6.0%)	13 (1.3%)	7.1 (0.4%)	7.9 (0%)
Commedia, Orlando Furioso (verse)	6.6–8.4 (3.8%)	4.4–4.7 (0–0.6%)	4.4–5.4 (0%)	2.4–3.9 (0%)

scale G is

$$r_G = \frac{\Pr(g \leq G)}{\Pr_{\text{unif}}(g \leq G)}. \quad (4)$$

Table 1 and Figure 3 give the result. In prose a word used twice in 38 k tokens is a topic word and its two uses sit in the same passage: $r_{100} = 12\text{--}50$, with 5–23 % of doubletons on one page. The manuscript’s doubletons show a tenth of that ($r_{100} = 3.8\text{--}3.9$, 2–3 % same page), below even Italian narrative verse (6.8–10, 4 %), whose rare forms are scattered by rhyme and metre; the excess within 10 tokens ($r_{10} = 4\text{--}8$) is the known line-level near-repeat phenomenon. The picture does not change with k : at every frequency class the manuscript clusters like rhymed verse, not like prose, in both Currier languages. [record: §2, 6.1]

The corpus sweep places the manuscript against 148 known windows on the pooled $k \in [2, 5]$ statistic: known r_{100} runs from 3.39 (5th percentile 4.66, median 12.9) to 34.0; [IT2a](#) scores 3.89 (2.5th percentile, log-z -2.3 ; below it only Tasso’s *Gerusalemme liberata*), [RF1b](#) 4.38 (3.8th percentile, log-z -2.1 ; below it Tasso and Dante).⁴ Currier *A* or *B* alone score 2.8–2.9, below every known window. The whole-manuscript stream mixes languages and sections, which adds clustering, so these are upper bounds on the within-section locality. [record: §9]

This agrees with the two published scales: section-level vocabulary information (Montemurro & Zanette) is present—adjacent-page sharing is $z = 6\text{--}7$ under a global page shuffle and mostly removed by permuting only within section [record: §3]—while at passage scale rare recurrences are nearly memoryless (Reddy & Knight; Schinner). Whatever a manuscript “word” is, its rare recurrences carry very little topical locality; every test below has to find an order signal in a text with roughly one tenth of the locality of the controls.

4 The leaf-pair test

4.1 Statistic and null

Let $\mathcal{P}$ be the set of occurrence pairs of rare types whose two members lie on different leaves of the *same quire*. Each pair is classified by the relation between its two leaves:

- $\mathcal{C}$ conjugate: $s(i) = s(j)$, $\ell(i) \neq \ell(j)$ —the two leaves of one sheet (one bifolium);
- $\mathcal{N}$ nested-adjacent: the leaves are consecutive in the binding and $s(i) \neq s(j)$;
- $\mathcal{B}$ the innermost sheet, whose two leaves are both conjugate and nested-adjacent (kept apart);

⁴The recorded percentiles were computed on the sweep log’s 158 windows, which include the ten Italian raw-text documents twice; on the 148 de-duplicated windows they are 1.4 % ([IT2a](#)) and 2.0 % ([RF1b](#)). Either way the manuscript sits at the extreme low edge of the corpus.

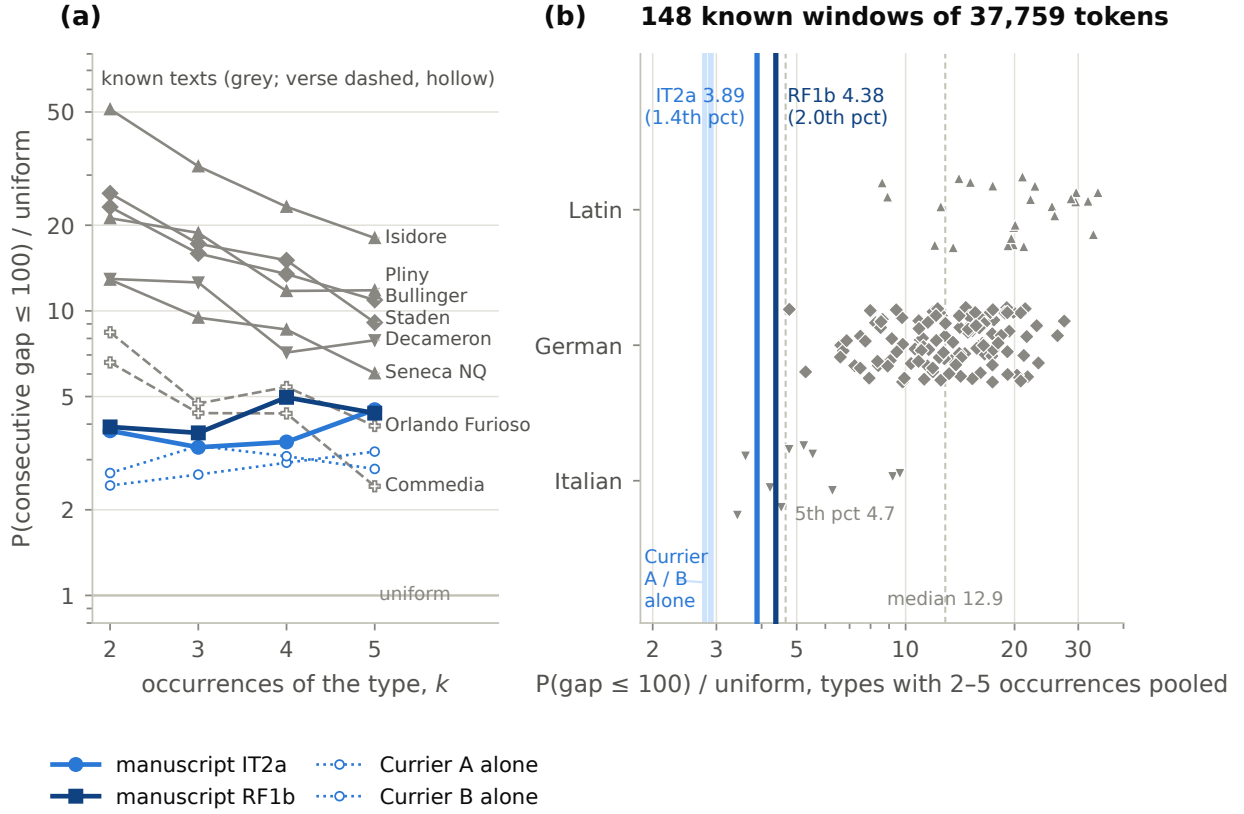


Figure 3. Rare-text locality. (a) r_{100} by frequency class k for the manuscript (whole stream and Currier *A/B* alone) against known prose and verse (Table 1). (b) Corpus sweep: 148 known 37,759-token windows on the pooled $k \in [2, 5]$ statistic, with the two manuscript transcriptions marked at their percentiles among these 148 windows (the recorded 2.5 % / 3.8 % were on the sweep log’s 158-window count; see the footnote in the text). [record: §6.1, 9]

$\mathcal{O}$ every other same-quire leaf pair.

The statistic is the difference of the two discriminating counts,

$$D = n_{\mathcal{C}} - n_{\mathcal{N}}, \quad n_X = \#\{(i, j) \in \mathcal{P} : (i, j) \in X\}. \quad (5)$$

The null permutes page *contents* among page *slots* within groups $g(p) = (q(p), \lambda(p))$ —quire and Currier language—or, for the hand control, $g(p) = (q(p), \lambda(p), h(p))$. A permutation π moves whole pages, so every page keeps its own rare-text load and its own within-page clumping; only which physical slot a page occupies is randomised. D_π is recomputed per draw (2,000 draws; 5,000 for the tail study, Section 4.5), giving z_D and Equation (3). Because the innermost sheet is excluded from both $\mathcal{C}$ and $\mathcal{N}$ and every quire with a single sheet contributes nothing, D is driven by quires with ≥ 2 sheets.

Under a text written in the nested order Equation (1), passage locality lands on $\mathcal{N}$ pairs and $D < 0$; under a text written in the stacked order Equation (2) it lands on $\mathcal{C}$ pairs and $D > 0$; under evenly spread rare text $D \approx 0$.

4.2 Power controls

The magnitude of D under either hypothesis depends on the text’s locality (Section 3), so it is calibrated with known texts laid onto the *manuscript’s own page slots*: the same tokens per page (symbols per page for n -grams), the same quire and sheet structure. Each text is written either in the nested order or in the stacked order, then bound nested and analysed identically.

Table 2. The leaf-pair statistic $D = n_C - n_N$ (Equation (5)) on the manuscript, as z against the within-quire-and-language page-permutation null, and the range of z_D the six control texts give on the same page slots when written nested or stacked. Word rows pool types with the stated k ; n -gram rows are $k \in [2, 10]$. [record: §4, 6.2, 7.1, 7.2]

unit	pairs (IT2a/RF1b)	manuscript z_D		controls z_D (six texts)	
		IT2a	RF1b	written nested	written stacked
words, $k = 2$	896 / 997	+0.7	−0.1	−1.8 to −4.0 [†]	+0.2 to +3.9 [†]
words, $k \in [2, 5]$	4,839 / 5,306	+2.5	+1.8	−0.8 to −7.1	+0.9 to +3.6
words, $k \in [2, 10]$	13,665 / 14,625	+2.4	+2.5	−2.0 to −7.3	+1.5 to +4.7
n -grams $n = 4$, all	33 k / 37 k	+2.6	+1.2	—	—
$n = 5$, all	74 k / 81 k	+3.4	+2.8	−2.2 to −6.3	+1.1 to +4.7
$n = 6$, all	113 k / 119 k	+3.3	+3.5	−3.4 to −7.6	+1.7 to +4.5
$n = 7$, all	119 k / 118 k	+4.1	+4.4	−3.7 to −7.8	+1.9 to +4.5
$n = 8$, all	98 k / 92 k	+4.7	+4.8	—	—
$n = 5$, cross		+3.1	+2.3	−1.7 to −7.0	+1.3 to +4.7
$n = 6$, cross		+3.2	+3.5	−2.4 to −7.2	+1.6 to +4.9
$n = 7$, cross		+3.7	+4.2	−2.4 to −7.3	+1.4 to +4.3
$n = 8$, cross		+4.4	+5.2	—	—

[†] five texts (no Commedia), 500 shuffles; verse written stacked is indistinguishable from the null at $k = 2$.

Six texts were used (Isidore, Seneca *NQ*, Bullinger, Decameron, Commedia, Orlando Furioso), covering 3–13× the manuscript’s locality.

4.3 Results

Doubletons alone cannot tell. With the 896 / 997 doubletons, D is −9 against a null of -13.3 ± 6.3 (IT2a, $z = +0.7$) and −14 against -13.4 ± 6.6 (RF1b, $z = -0.1$). The controls show why: even texts with 3–13× the manuscript’s locality separate the two hypotheses by only 2–7 sd at $k = 2$, and stacked-written verse is indistinguishable from the null. The manuscript’s value lies above every nested-written control and inside the stacked-written band, but its expected magnitude under *either* hypothesis is smaller than any control’s. [record: §4]

Pooling rare types. Pooling all occurrence pairs of types with $k \in [2, 10]$ raises the pair count to 13.7 k / 14.6 k and puts the manuscript at $z_D = +2.4 / +2.5$ —inside the stacked-written band (+0.9 to +4.7) and outside the nested-written band (−0.8 to −7.3, all six texts, both pool sizes). Both components move: conjugate leaves share *more* than random same-quire leaves ($z = +2.2 / +2.3$) and nested-adjacent leaves share *less* ($z = -1.8 / -1.8$). The excess is spread over quires A (herbal, Currier A), T (stars, B), H and M and absent from herbal quires B–G; Currier B pages alone give +1.8 / +1.9, Currier A pages alone +0.3 / +0.7. This pooled statistic was chosen *after* the $k = 2$ test came back null, so its nominal one-sided $p \approx 0.01$ is not pre-registered. [record: §6.2]

Glyph n -grams: the pre-stated confirmation. An independent unit was run as the check of the pooled-word hypothesis: character n -grams, $n = 4–8$, $k \in [2, 10]$, with the cross variant to exclude fragments of single rare words. The excess grows with n from +2.6 / +1.2 at $n = 4$ to +4.7 / +4.8 at $n = 8$ (cross +4.4 / +5.2); doubleton n -grams alone give +4.5 / +3.4 at $n = 7$. On the controls, all 36 nested-written cells (six texts $\times n = 5, 6, 7 \times$ all/cross) are negative (−1.7 to −7.8) and all 36 stacked-written cells positive (+1.1 to +4.9); the manuscript’s +3 to +5 sits in the upper part of the stacked-written band (Figure 4, Figure 5). [record: §7.1, 7.2]

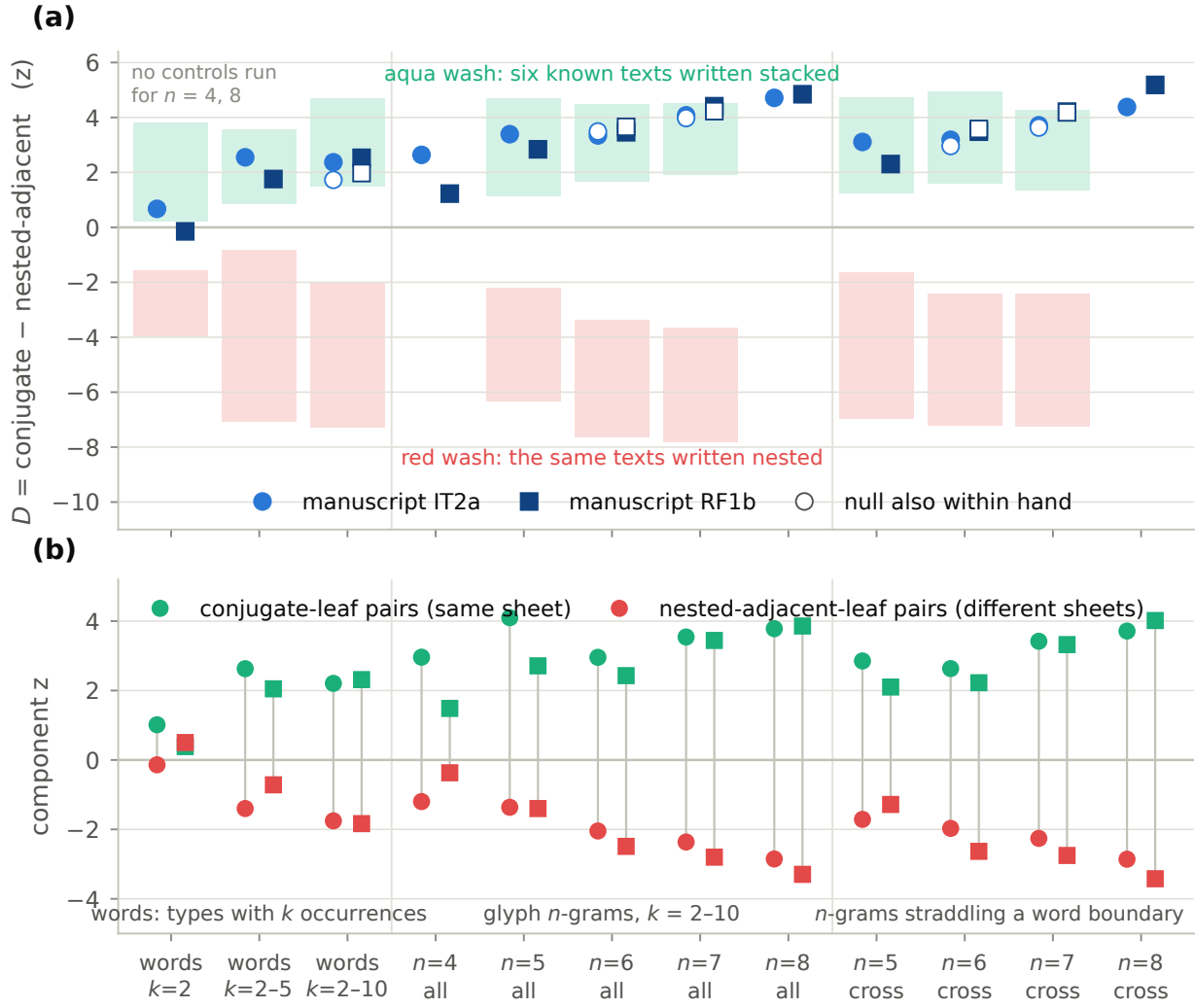


Figure 4. The leaf-pair statistic on the manuscript. (a) z_D by unit (words at three pool sizes; n -grams $n = 4-8$, all and boundary-straddling) for both transcriptions, with the control bands: the range of z_D six known texts give on the same page slots when written nested (red) or stacked (aqua); hollow markers are the hand-controlled null. (b) The two components, conjugate and nested-adjacent, separately. [record: §4, 6.2, 7]

4.4 Scribal hand

Conjugate leaves are in the same hand (and language) for 97 % of page pairs (138/142 IT2a, 146/152 RF1b); nested-adjacent leaves only 70 % (203/290, 203/292) and other same-quire pairs 73–75 %. The hands run by sheet—itself a codicological observation consistent with sheets being the working unit—so a hand-specific spelling habit could make conjugate leaves share rare n -grams for reasons unrelated to reading order. Restricting the null to permutations within quire, language *and* hand (28 / 30 groups) removes that route (Table 3): the n -gram result is unchanged and the word result drops by about a third but keeps its sign. [record: §7.3]

4.5 How often does even spreading produce the excess?

The z values assume nothing about the null's shape, but a tail of a few in 10^5 should be counted, not extrapolated. Table 4 gives, for 5,000 draws per cell, the number of null draws reaching the observed D under two readings of “evenly spread”: *perm*, the page-content permutation above (the conservative reading, since page-level clumping stays in the variance),

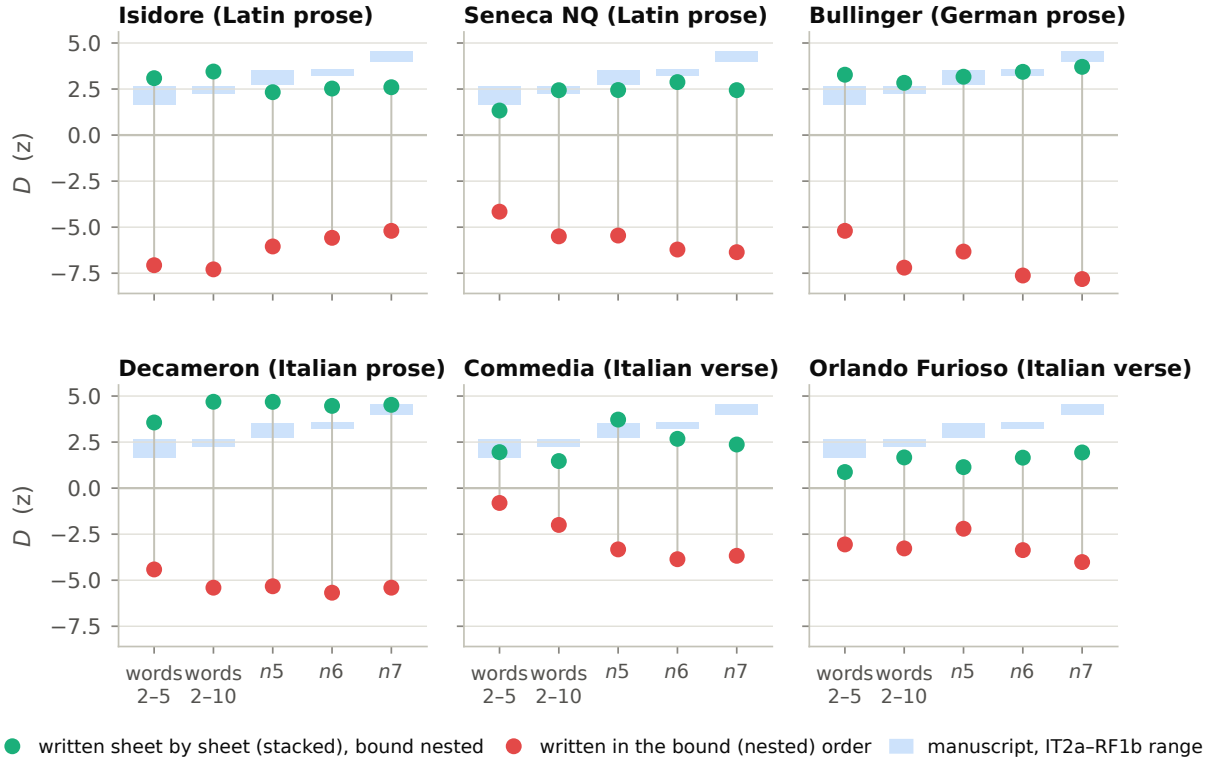


Figure 5. Power controls, per text: z_D when the text is written in the nested order (red) and in the stacked order (aqua) onto the manuscript’s page slots, for words $k \in [2, 10]$ and n -grams $n = 5, 6, 7$. Every nested-written cell is negative and every stacked-written cell positive; the manuscript’s values are the reference lines. [record: §6.2, 7.2]

Table 3. Hand control: z_D (IT2a / RF1b) when page contents are permuted within quire and language, and within quire, language and hand. [record: §7.3]

unit ($k \in [2, 10]$)	null: quire + language	null: quire + language + hand
words	+2.4 / +2.5	+1.7 / +2.0
$n = 6$ all ; cross	+3.4 / +3.3 ; +2.9 / +3.2	+3.5 / +3.7 ; +3.0 / +3.6
$n = 7$ all ; cross	+3.9 / +4.1 ; +3.6 / +4.1	+4.0 / +4.2 ; +3.6 / +4.2

and *uniform*, every occurrence of a rare type landing independently on a page of its language with probability proportional to page length. [record: §12]

The word-level excess alone is a 1-in-30 to 1-in-200 event under even spreading—real but not decisive, and chosen after a null result. The n -gram excess, the pre-stated confirmation, never occurs in 5,000 even-spread draws at $n \geq 7$ on either transcription or either null; its normal-tail chance is of order 10^{-5} – 10^{-6} per cell. The cells are not independent (n -grams of neighbouring n overlap and the two transcriptions are one text), so the family is best counted as one test per transcription at $n = 7$ or 8; even a twenty-fold multiplicity correction leaves it below 1/1,000 (Figure 6).

Table 4. Empirical tail of D : null draws (of 5,000) reaching the observed value, IT2a / RF1b, with the normal-tail p for the recorded z . Under the uniform null every n -gram cell is 0/5,000 ($z = +3.6$ to $+8.2$) and the word cell 10 / 0. [record: §12]

unit ($k \in [2, 10]$)	perm within quire + language			perm within quire + language + hand		
	draws $\geq$ obs	z	normal tail	draws $\geq$ obs	z	normal tail
words	54 / 26	+2.3 / +2.5	$1/90 \cdot 1/190$	157 / 57	+1.8 / +2.0	$1/30 \cdot 1/90$
$n = 5$	1 / 14	+3.5 / +2.9		2 / 2	+3.5 / +3.3	
$n = 6$	3 / 3	+3.5 / +3.5	$\approx 1/1,700$	5 / 1	+3.6 / +3.7	
$n = 7$	0 / 0	+4.2 / +4.3	$\approx 1/70,000$	0 / 0	+4.1 / +4.3	
$n = 8$	0 / 0	+4.6 / +5.0	$\approx 1/500,000$	0 / 0	+4.5 / +4.9	

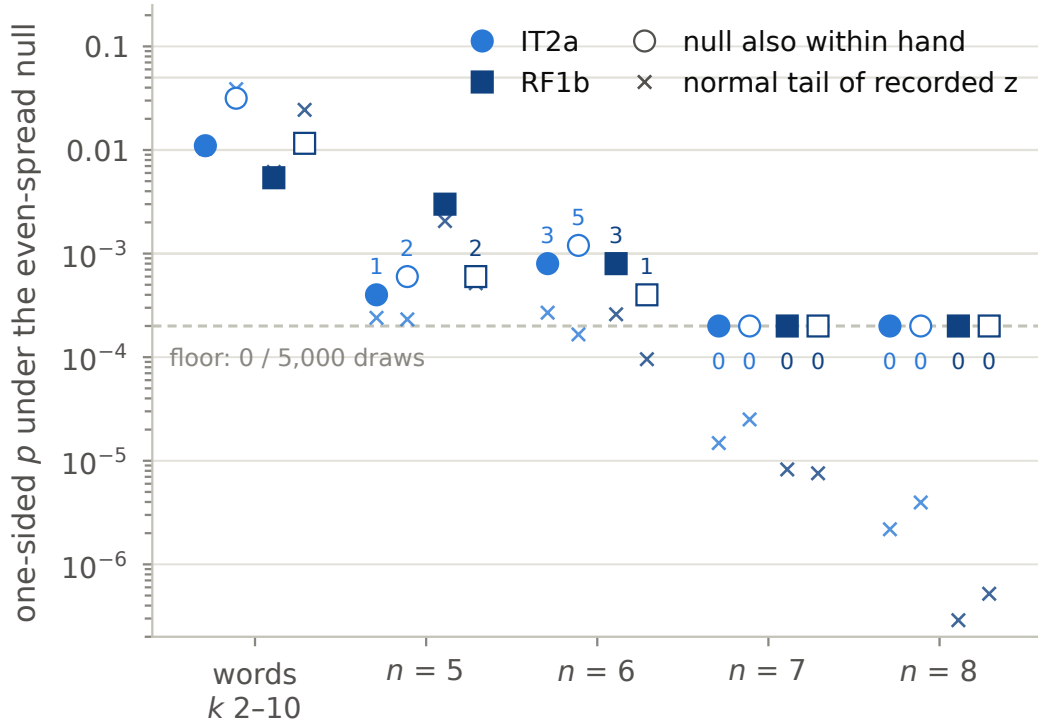


Figure 6. Empirical tail of the leaf-pair excess: null draws out of 5,000 reaching the observed D , by unit, null and transcription; zero counts are shown explicitly. [record: §12]

What this establishes

Text on the two leaves of one physical sheet shares its rare text more than text on leaves that are neighbours only in the binding, and the nested neighbours share *less* than random same-quire leaves. Three units (words, within-word n -grams, boundary-straddling n -grams), two transcriptions and a null holding quire, Currier language and hand fixed agree; six real texts written in the binding order never show this (36/36 cells negative) and written sheet by sheet always do (36/36 positive) at the manuscript's magnitude. Under even spreading the $n \geq 7$ excess is a $\lesssim 10^{-4}$ event.

5 What the sheet effect is and is not

5.1 Rare text is not contained by sheet

The sheet effect could mean that rare types are *held* by one sheet (one bifolium). They are not (Table 5, Figure 7a). Three quarters of doubletons and over 90% of the 3–5-times types

Table 5. Where the occurrences of a rare type sit: percentage of types with all occurrences in the given unit, [IT2a](#) / [RF1b](#) (null in brackets: page contents shuffled within quire and language, 200 draws; same-page and 2⁺-quire columns are order-invariant under that null). [record: §11]

<i>k</i>	types	same page	same leaf (<i>r</i> + <i>v</i>)	same sheet, both leaves	2 ⁺ sheets, same quire	2 ⁺ quires
2	896 / 997	2.9 / 2.3	2.7 / 2.6 (1.6 / 1.8)	2.6 / 2.8 (2.3 / 2.6)	14.3 / 14.4 (15.6 / 15.5)	77.6 / 77.8
3	399 / 413	0.3 / 0.2	0.3 / 0.5 (0.2 / 0.1)	1.0 / 0.2 (0.4 / 0.3)	7.3 / 8.0 (8.0 / 8.4)	91.2 / 91.0
4	216 / 250	0.0 / 0.8	0.0 / 0.0	0.0 / 0.4 (0.1 / 0.1)	7.4 / 6.8 (7.4 / 7.0)	92.6 / 92.0
5	145 / 157	0.0 / 0.6	0.0 / 0.0	0.0 / 0.6 (0.0 / 0.1)	3.4 / 3.8 (3.4 / 4.3)	96.6 / 94.9

span more than one quire; only 2–3 % of doubletons and ≈ 0 –1 % of the rest fall within a single sheet. The Section 4 effect is a small excess in the roughly 20 % of pairs that fall within a quire (same-leaf 2.7 % vs 1.6 % expected; same-sheet 2.6 % vs 2.3 %). This is why the two re-ordering exercises below move the locality statistics so little: the material an order can act on is a fifth of the pairs. [record: §11]

5.2 The stacked order does not change solver windows

The decipherment programme’s word-homophonic solver has a findability wall near 4 tokens per type and had rejected windowing at 1.8–2.0 tokens/type per 1,024-token window. Re-reading the Currier *A* and *B* streams in the stacked order changes tokens/type in non-overlapping windows by ≤ 0.01 (under 1 %) at $W \leq 1,024$ —occasionally detectable against the within-quire page-shuffle null (+3.2 sd for *A* at $W = 512$ on [IT2a](#) and *B* at $W = 1,024$ on [RF1b](#))—and by nothing or slightly negatively at $W \geq 2,048$; every window stays at 1.5–3.0 tokens/type with 70–80 % hapax types (Figure 7b). The order result opens no solver path. [record: §8]

5.3 No sheet order lifts the locality above the noise ceiling

If the sheets were misarranged, a better stacking order should raise the manuscript’s rare-text locality toward the corpus. Simulated annealing (20,000 steps) over sheet orders was run on [IT2a](#) (52 sheets, single surviving leaves counted as sheets, every sheet read stacked) with objective $J(\sigma) = \sum_{(i,j)} \exp(-d_\sigma(i,j)/1000)$ over consecutive occurrences of types with $k \in [2, 5]$, under four constraint levels—*strict* (swaps only within the same Currier language, section and hand; 12 groups), *topic* (language + section; 10), *language* (3), *free*—and reported as r_{100} and r_{1000} (Equation (4)), whose corpus percentiles are known from Section 3 (r_{1000} : 5th percentile 1.65, 25th 2.74, median 3.35). [record: §10]

Four readings of Table 6 and Figure 8. (i) Re-ordering the sheets under any constraint moves r_{1000} from 2.09 to 2.4–2.6, but a manuscript whose page contents were first shuffled—so that no order information exists—is moved by the same optimiser to 2.3–3.0: the whole gain is what an optimiser extracts from noise on $\sim 5,000$ rare pairs. (ii) Random sheet orders are optimised to the same values as the actual stacked order; the current order is not a special optimum. (iii) Known texts written sheet by sheet and shuffled are put back to their true-order value plus 0.0–0.35 of over-fitting, so the method would recover a real order if one carried locality. (iv) Even the over-fitted manuscript values stay at the bottom of the corpus (r_{1000} 2.4–2.6 is below the 25th percentile), and r_{100} barely moves because gaps under 100 tokens fall inside a sheet (≈ 670 tokens). Allowing inverted folding (b^v, b^r, a^r, a^v) adds $\approx +0.05$ to both the manuscript and the noise runs. Sheet order is not what separates the manuscript’s rare text from prose; the separation is within the sheet. [record: §10]

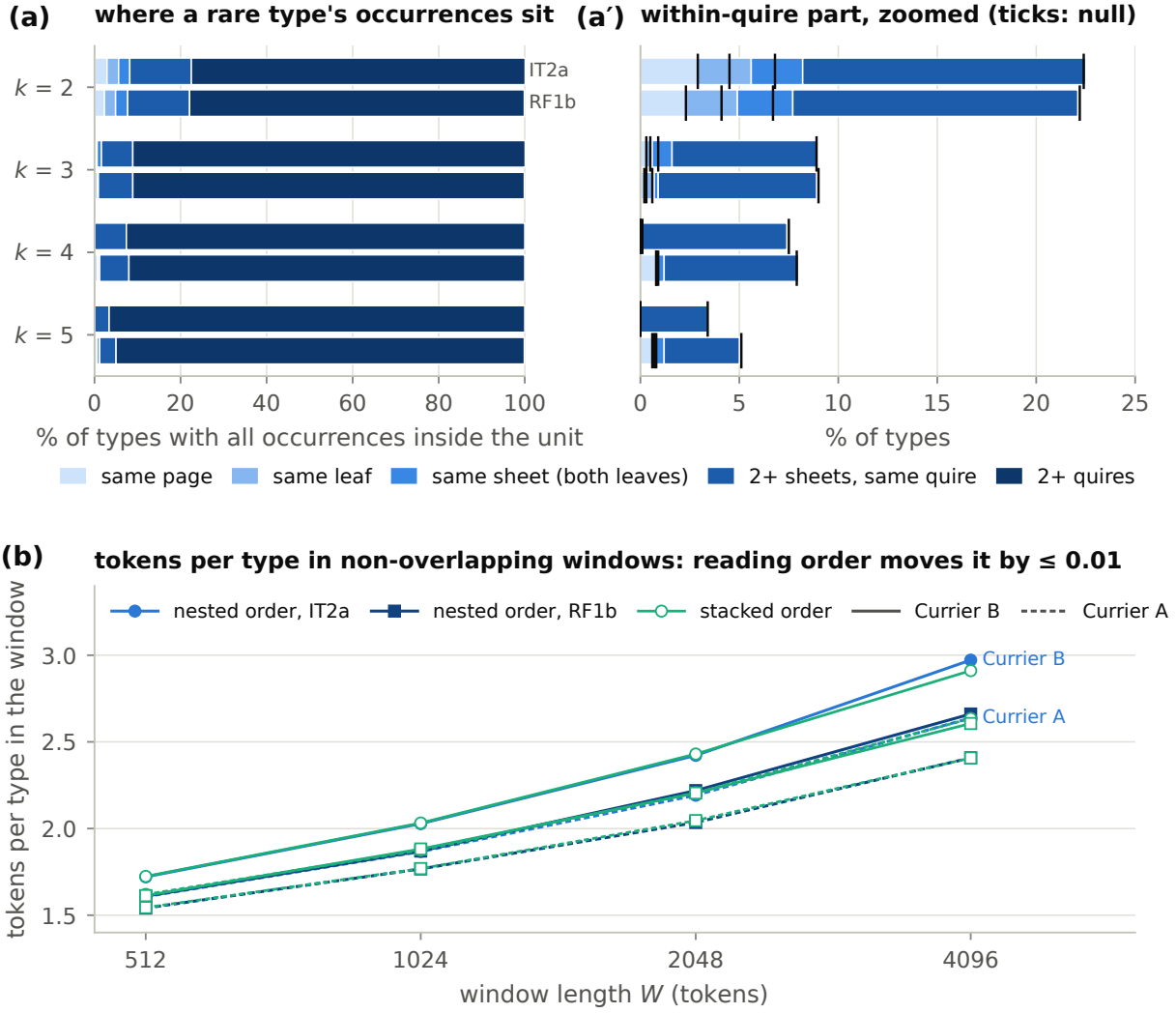


Figure 7. (a) Containment of rare types by codicological unit, Table 5. (b) Tokens per type in non-overlapping windows of the Currier A and B streams read nested versus stacked. [record: §11, 8]

6 Seriation within a quire

The leaf-pair test says the sheet (bifolium) is a unit. Can the rare text also say *which sheet follows which*? For a quire of S sheets there are $S!$ stacked orders (each sheet read a^r, a^v, b^r, b^v), plus the nested binding as one more candidate (Section 7 widens the set to every nesting pattern of the sheets). Quires that are one language, one hand and one section throughout make the cleanest test: **M** (f75–f84: 20 pages, 5 sheets, Currier B, hand 2, 6,911 / 6,969 tokens; 120 orders) and **T** (f103–f116: 23 pages, 6 sheets, Currier B, hand 3, 10,673 tokens; 720 orders); the Currier-A herbal quires **A** (f1–f8, 1,495 tokens), **B** (f9–f16, three sheets and the single leaf f13, 1,019 tokens) and **C** (f17–f24, 1,401 tokens), hand 1, have 4 sheets and 24 orders. Sheets are numbered outer→inner, so the current stacked order is 1–2–...– S . [record: §13, 15]

6.1 Cost functions

Units are word types and glyph n -grams ($n = 5–8$) with 2–10 occurrences *in the quire*. Because the four pages of a sheet are contiguous in every stacked order, within-sheet pairs are the same for every candidate and would only add the Section 4 effect as a constant; the “which stacking” question therefore uses only between-sheet pairs, and the “stacked at all vs as bound”

Table 6. Sheet-order optimisation, orientation fixed: r_{100} / r_{1000} after annealing under each constraint level. The page-content-shuffled manuscript has no order information and shows the optimiser’s noise ceiling; known texts written stacked and shuffled are returned to their true-order value.
[record: §10]

start	stat.	init	strict	topic	language	free
manuscript, stacked order	r_{100}	4.04	4.24	4.28	4.24	4.48
	r_{1000}	2.09	2.40	2.45	2.50	2.62
manuscript, random sheet orders (3 starts)	r_{100}	4.04	4.2	4.1–4.2	4.2–4.4	4.3–4.4
	r_{1000}	2.09	2.42–2.45	2.43–2.47	2.42–2.51	2.58–2.68
manuscript, page contents shuffled first (3 draws)	r_{100}	4.0–4.5	5.3–5.7	5.1–5.4	5.3–5.9	6.1–6.6
	r_{1000}	1.89–1.97	2.32–2.54	2.37–2.59	2.51–2.83	2.67–3.00
<i>known texts written stacked, sheets shuffled; “init” column gives the true-order value</i>						
Isidore	r_{100}	29.3	29.2	29.3	29.2	29.2
	r_{1000}	5.09	5.09	5.11	5.09	5.16
Seneca NQ	r_{100}	8.69	8.64	8.69	8.59	8.77
	r_{1000}	2.51	2.52	2.56	2.61	2.64
Bullinger	r_{100}	14.8	14.6	14.7	14.5	14.8
	r_{1000}	3.36	3.46	3.46	3.50	3.55
Decameron	r_{100}	9.27	9.30	9.20	9.30	9.30
	r_{1000}	2.81	2.95	2.94	3.07	3.16
Orlando Furioso	r_{100}	5.36	5.56	5.43	5.52	5.49
	r_{1000}	1.96	2.07	2.10	2.21	2.21

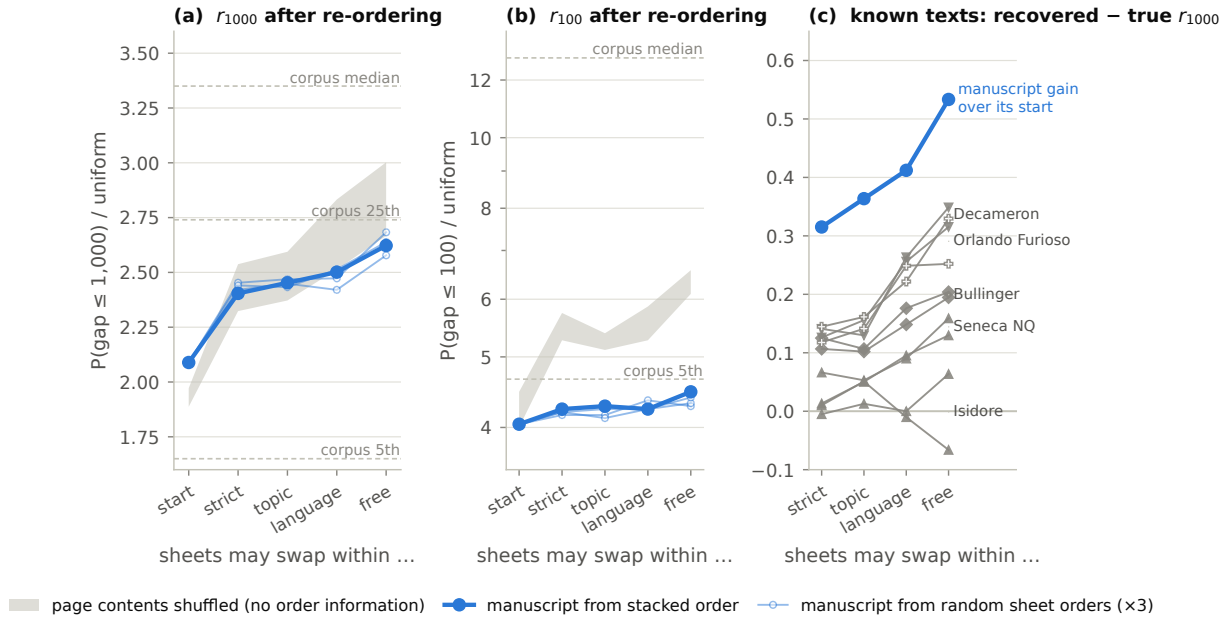
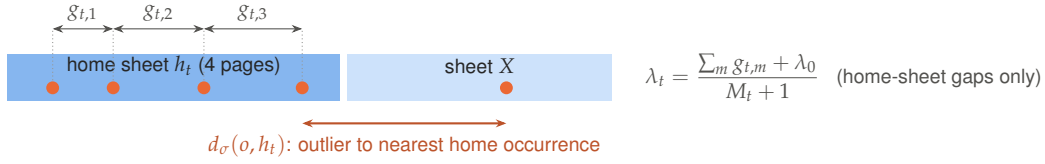


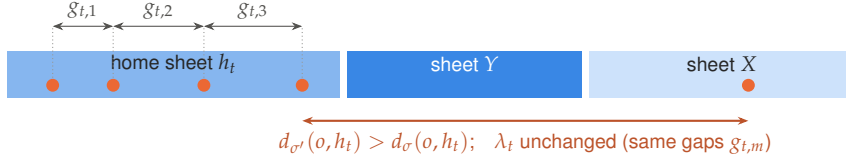
Figure 8. Sheet-order optimisation. r_{1000} after annealing under each constraint level for the manuscript from its stacked order and from random sheet orders, for the page-content-shuffled manuscript (noise ceiling), and for five known texts recovering their true-order value; corpus percentiles as reference lines. [record: §10]

question all cross-page pairs. Four costs are compared, each to be minimised over candidates σ :

candidate order $\sigma: \dots H X \dots$ (sheet X read directly after the home sheet)



candidate order $\sigma': \dots H Y X \dots$ (another sheet placed between them)



cost contribution of type t for the outlier o : cluster = d/λ_t , blog = $\log(1 + d/\lambda_t)$, modal = $w_t d$; L1 instead averages every cross-sheet pair distance.

Figure 9. The cluster-scaled seriation cost. A rare type’s home sheet holds most of its occurrences (orange); the gaps between them fix its home gap λ_t Equation (10), which is identical under every stacked order because a sheet’s four pages are always contiguous. Each occurrence outside the home sheet contributes its distance to the nearest home occurrence, measured in units of λ_t Equation (7); the candidate order changes d , never λ_t .

[record: §13, 14]

$$\text{L1: } c_{\text{L1}}(\sigma) = \frac{1}{|\mathcal{P}_{\text{between}}|} \sum_{(i,j) \in \mathcal{P}_{\text{between}}} d_\sigma(i, j), \quad (6)$$

$$\text{cluster: } c_b(\sigma) = \sum_t \sum_{o \in O_t \setminus h_t} \frac{d_\sigma(o, h_t)}{\lambda_t}, \quad d_\sigma(o, h_t) = \min_{o' \in O_t \cap h_t} d_\sigma(o, o'), \quad (7)$$

$$\text{blog: } c_{\text{bl}}(\sigma) = \sum_t \sum_{o \in O_t \setminus h_t} \log\left(1 + \frac{d_\sigma(o, h_t)}{\lambda_t}\right), \quad (8)$$

$$\text{modal: } c_m(\sigma) = \sum_t w_t \sum_{o \in O_t \setminus h_t} d_\sigma(o, h_t), \quad w_t = \frac{\max_p |O_t \cap h_t \cap p|}{|O_t \cap h_t|}. \quad (9)$$

Here h_t is the *home sheet* of type t , the sheet holding most of its occurrences; types with a tie (a doubleton split 1–1) carry no intra-sheet information and are skipped. The plain mean Equation (6) gives every cross-sheet pair the same weight and a k -occurrence type $\sim k^2$ pairs. The cluster cost Equation (7) is the negative log-likelihood of an exponential-tail cluster model with one term per *outlier* occurrence: a type that clusters tightly where it lives carries more positional information than one spread over its sheet. Its scale is set from the home sheet alone,

$$\lambda_t = \max\left(30, \frac{\sum_{m=1}^{M_t} g_{t,m} + \lambda_0}{M_t + 1}\right), \quad \lambda_0 = \text{median}_t\left(\frac{1}{M_t} \sum_m g_{t,m}\right), \quad (10)$$

with $g_{t,m}$ the gaps between consecutive home-sheet occurrences (shrunk toward the median over types by one pseudo-gap and floored at 30 symbols). Because within-sheet gaps are identical under every stacked order, λ_t is the same for all candidates and no candidate can tune its own weights (Figure 9). The log form Equation (8) caps a single far outlier; the modal form Equation (9) is the discrete version of the same intuition.

6.2 Candidates, nulls and the best-of- N heuristic

For each unit and cost the candidate vector $\mathbf{c} = (c(\sigma))_\sigma$ over all $S!$ stacked orders is standardised within itself, $z_\sigma = (c(\sigma) - \bar{c})/s_c$, and the winner is $\sigma^* = \arg \min_\sigma c(\sigma)$ with

$z^* = z_{\sigma^*}$. Three nulls are attached to it. [record: §13–15]

- **Content shuffle.** Page contents are permuted among the quire’s page slots (200 draws) and the best of all candidates is recomputed each time; p_{best} is the empirical fraction Equation (3) of shuffles whose winner is at least as extreme (in z^*) as the real winner. Page-level clumping is retained by construction.
- **Geometry.** Page lengths stay in their slots and every occurrence of a rare type is re-placed uniformly over the quire’s tokens (200 draws): does sheet *size* alone favour the winning order? Valid for L1 and modal, which are means of distances; *not* for cluster and blog, whose λ_t changes under uniform placement (their geometry p is ≈ 1 everywhere and is discarded).
- **Best-of- N .** The minimum of N roughly normal candidate values is expected near $-\Phi^{-1}(1 - \frac{1}{N+1})$: about -2.4sd for $N = 120$, -3.0 for 720, -1.75 for 24. A winner at that depth stands out from *nothing*; the content-shuffle winners give the empirical version (Section 9).

Cross-unit consistency is the correlation of the candidate vectors of two units (e.g. words vs $n = 7$) against the same correlation on shuffled contents, which keeps the overlap of n -grams with their words.

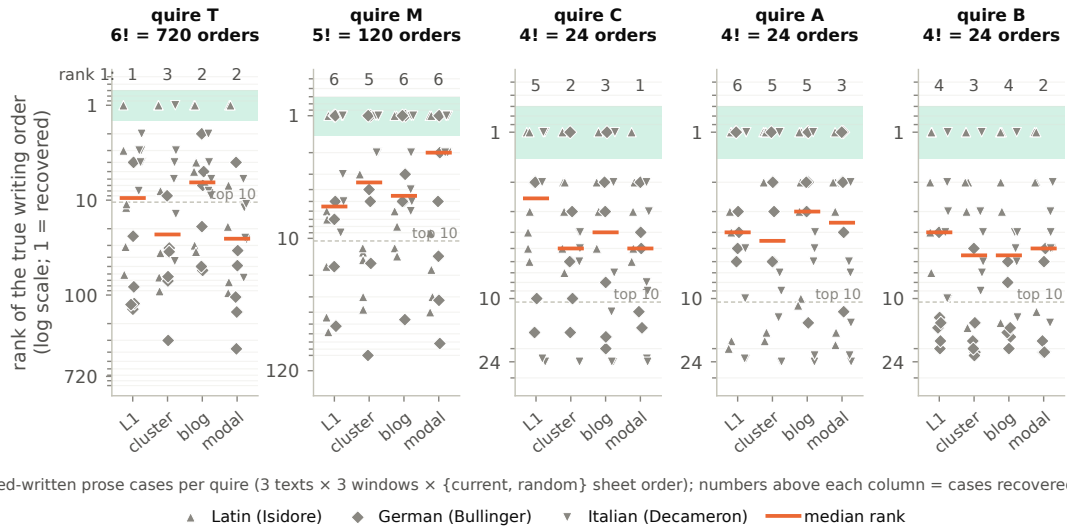
Power. Known prose was written onto each quire’s page slots, stacked in the current or a random sheet order, or nested (three texts $\times$ three windows). On quire M’s slots the true stacked order ranks 1/120 in 4–6 of 18 cases and in the top 10 in 13 under L1 (median rank 6); the cluster-scaled costs lower the median rank to 2–4 (cluster 3/13, blog 6/13, modal 5/13 for rank-1/top-10). Written nested, the nested order ranks 1/121 in 8 of 9. Across the five quires, rank-1 counts (L1 / blog) are 6/6 (M), 1/2 (T; median rank 10/6 among 720), 6/5 (A), 4/4 (B), 5/3 (C), with top-10 in 13–15 of 18 on the four-to-five-sheet quires and 9–13 on T. So the metric recovers a prose writing order outright about a third of the time and usually to within the top tenth; with the manuscript’s locality a tenth of prose (Section 3), low power there is expected (Figure 10). A **nested-written control** asks which *stacked* order prose written as bound prefers (L1, nine nested-written cases per quire): the bound order 1–2–...–S exactly in T 2/9, M 1/9, C 5/9, A 4/9, B 5/9 cases; an order within three pairwise inversions of it in T 6/9, M 8/9, C 9/9, A 9/9, B 9/9; an order with the outermost sheet adjacent to the innermost in T 1/9, M 3/9, C 4/9, A 4/9, B 0/9. So nested writing pulls the stacked seriation toward the bound order or a near variant, and produces the outermost-next-to-innermost feature at or below its chance rate (Section 6.3). [record: §13, 14, 15]

6.3 Results

Stacked at all versus as bound. On quire M, with all cross-page pairs, the nested binding ranks 100–121 of 121 on every unit and transcription (121/121 for $n = 5, 8$ on IT2a and $n = 7, 8$ on RF1b) and is no better than random page contents ($z = -0.6$ to $+0.9$), while the stacked candidates as a set beat shuffled contents at $z = +2.3$ to $+6.4$ —the Section 4 result in seriation form: any reading that keeps a sheet’s four pages together beats the binding, and the binding is as bad as chance. [record: §13]

Which stacking. Table 7 summarises the between-sheet seriation on the five quires; Figure 12 details quire T and Figure 13 the p -grid. [record: §14, 15]

Quire T is unambiguous by this test. Every n -gram length, both transcriptions and all four costs choose 1–6–5–4–2–3 (words: 1–6–5–4–3–2); it beats shuffled contents in ≤ 1 of 200 draws on every cell, beats sheet geometry likewise, sits 3–4sd below the candidate mean where a random minimum of the candidate set would sit near 2.9, and the current outer→inner order lies in the bottom half to bottom tenth of the 720. In folios: f103/f116, f108/f111, f107/f112, f106/f113, f104/f115, f105/f114. [record: §15]



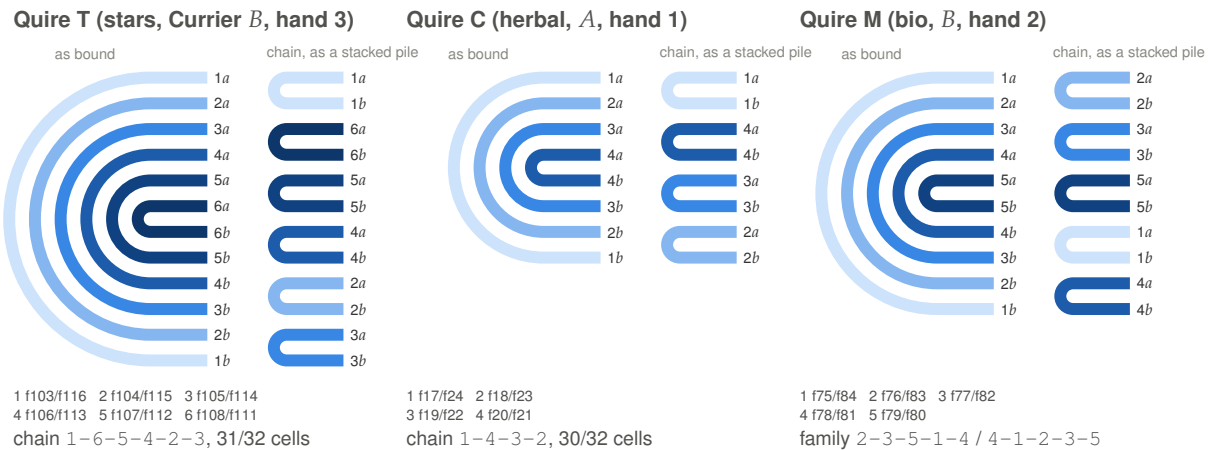
18 stacked-written prose cases per quire (3 texts $\times$ 3 windows $\times$ {current, random} sheet order); numbers above each column = cases recovered at rank 1

Figure 10. Seriation power on known prose written onto each quire’s page slots: rank of the true writing order among the candidate stacked orders (18 stacked-written cases per quire) under the four costs, and rank of the nested order among all candidates for nested-written prose. Nested-written control (L1, 9 cases per quire), best stacked order = the bound order exactly: T 2/9, M 1/9, C 5/9, A 4/9, B 5/9; within three pairwise inversions of it: T 6/9, M 8/9, C 9/9, A 9/9, B 9/9; outermost sheet adjacent to innermost: T 1/9, M 3/9, C 4/9, A 4/9, B 0/9. [record: §13–15]

Table 7. Between-sheet seriation per quire (both transcriptions; n -gram units $n = 5$ –8 unless stated). Orders list sheets outer $\rightarrow$ inner by number; “bound-order rank” is the rank of the current outer $\rightarrow$ inner order 1–2–...– S among the $S!$ candidates. The reversal-equivalence used when these were first recorded is corrected in Section 8, which leaves the adjacency (which sheets neighbour which) unaffected. [record: §14, 15]

quire (sheets, orders)	n -gram best order	p_{best} , content shuffle	z^* within candi- dates	p , geometry	bound- order rank	words (order; p)
T (6, 720)	1–6–5–4–2–3 31/32 cells ^a	L1 0.005–0.02 blog 0.005–0.02 modal 0.005–0.02	–3.1 to –4.05	L1 0.005 modal 0.005–0.02	150–626	1–6–5–4–3–2 (IT2a) 1–6–5–4–2–3 (RF1b) L1 0.005; modal 0.01–0.015; cluster, blog 0.36–0.74
M (5, 120)	2–3–5–1–4 3–2–5–1–4 4–1–2–3–5 one family ^b	L1 0.005–0.07 blog 0.01–0.12 modal 0.005–0.03	–2.3 to –3.2	L1 0.005–0.01 modal 0.005–0.03	30–114	2–3–5–1–4 0.03–0.4
C (4, 24)	1–4–3–2 30/32 cells	L1 0.005–0.15 blog 0.01–0.07 modal 0.005–0.27	–1.2 to –2.1	L1 0.005 modal 0.005–0.07	4–10	2–3–1–4, 1–2–3–4, 2–1–3–4 0.03–0.7
A (4, 24)	split ^c 1–2–4–3 1–4–2–3 3–1–2–4	L1 0.01–0.1 blog 0.035–0.26 modal 0.02–0.4	–1.4 to –2.2	L1 0.005 modal 0.01–0.9	10–20	2–4–1–3 0.1–0.5
B (3 sheets + leaf f13, 24)	1–2–3–4, the bound order ($n = 6$ –8)	L1 0.18–0.65 blog 0.06–0.56 modal 0.11–0.75	–1.3 to –2.4	L1 0.005–0.17 modal 0.005–0.68	1–4	1–4–3–2 and others 0.1–0.8

^a The single exception is IT2a $n = 5$ under L1, cluster and modal, which give 6–1–5–4–2–3 (sheets 1 and 6 swapped at the head of the chain). ^b One family: sheets 2 and 3 (f76/f83, f77/f82) together, then 5 (f79/f80), then 1 (f75/f84), with 4 (f78/f81) at an end; 4–1–2–3–5 is the same family with sheet 1 moved next to sheet 4. ^c 1–2–4–3 under L1 and cluster at $n = 5$ –7; 1–4–2–3 at $n = 7$ –8; 3–1–2–4 under blog and modal at $n = 5$ –6.



Chains are **adjacency only**: which sheet neighbours which. The sense of reading along a chain (top→bottom or bottom→top) is *not* established (Section 8).

Pattern shared by T, C, M: the outermost sheet (1) is adjacent to the innermost sheet (S), and the remaining sheets run outward in descending order (M with sheet 4 displaced to an end).

Figure 11. Sheet-adjacency chains selected by the seriation in quires T, C and M, drawn as in Figure 1. Left of each panel: the quire’s sheets as bound (1 outermost); right: the same sheets each folded on their own and piled in the selected chain order, colour-coded by physical position. The outermost-next-to-innermost feature has chance rate $2/S$ under a random order ($1/3$ for T, $1/2$ for C), so it is informative for T and at chance for the four-sheet quires on its own; C’s replication is the full chain 1-4-3-2. [record: §15]

Quire C replicates the shape with a smaller quire in a different language, hand and section: 1-4-3-2 on 30 of 32 cells, p 0.005–0.05 on most, geometry excluded. [record: §15]

Quire M is consistent. The plain mean (L1) found the best of 120 orders exactly as deep as the best of random values and a different optimum per unit [record: §13]; the cluster-scaled costs make the n -gram units agree on one family—sheets 2 and 3 (f76/f83, f77/f82) together, then 5 (f79/f80), then 1 (f75/f84), then 4 (f78/f81) at the end, or 4-1-2-3-5 with 1 moved next to 4—with the best-of-120 beating shuffled contents at p 0.005–0.05 in 14 of 16 n -gram $\times$ cost $\times$ transcription cells (modal 8/8 at ≤ 0.035). The words stay weak (p 0.1–0.4). The between-sheet affinity matrix (shared rare pairs, observed/expected) reads the same way: sheets 1 and 5 share most (1.35–1.42 words, 1.04–1.08 at $n = 7$), 2-3, 3-5 and 2-5 share 1.1–1.2, and sheet 4 shares least with everything (0.63–0.87, except 1.0 with sheet 1). [record: §13, 14]

Quire A shows sharing above shuffled contents (L1 p 0.01–0.05, geometry excluded) but no single order; the current order is disfavoured. **Quire B** (three sheets and a leaf, 1,000 tokens) prefers its current order but not above shuffled contents: no information. [record: §15]

The pattern across T, C and M. The outermost sheet sits next to the innermost sheet, and the remaining sheets then run outward in descending order: T 1-6-5-4-2-3 (n -grams; words 1-6-5-4-3-2), C 1-4-3-2, M 4-1-5-3-2 (words, $n = 8$; the sheet-reversal of the 2-3-5-1-4 of Table 7, written this way round to show the pattern) with sheet 4 displaced to the front (Figure 11). Two chance rates qualify the pattern. Under a uniformly random sheet order the outermost and innermost sheets are adjacent with probability $2/S$: $1/3$ for T ($S = 6$) and $1/2$ for the four-sheet quires C and A. The feature is therefore informative for T—where nested-written prose shows it in 1 of 9 cases and the bound order is at chance—and, taken on its own, at chance for C and A; what C replicates is a specific full chain (1-4-3-2 on 30 of 32 cells), not this feature. Nested-written prose does not produce the chains (its best stacked order is the bound order or a near variant, Section 6.2), and neither sheet geometry nor page-level clustering does (the content shuffle keeps each page’s own clusters). What could: composition sheet by

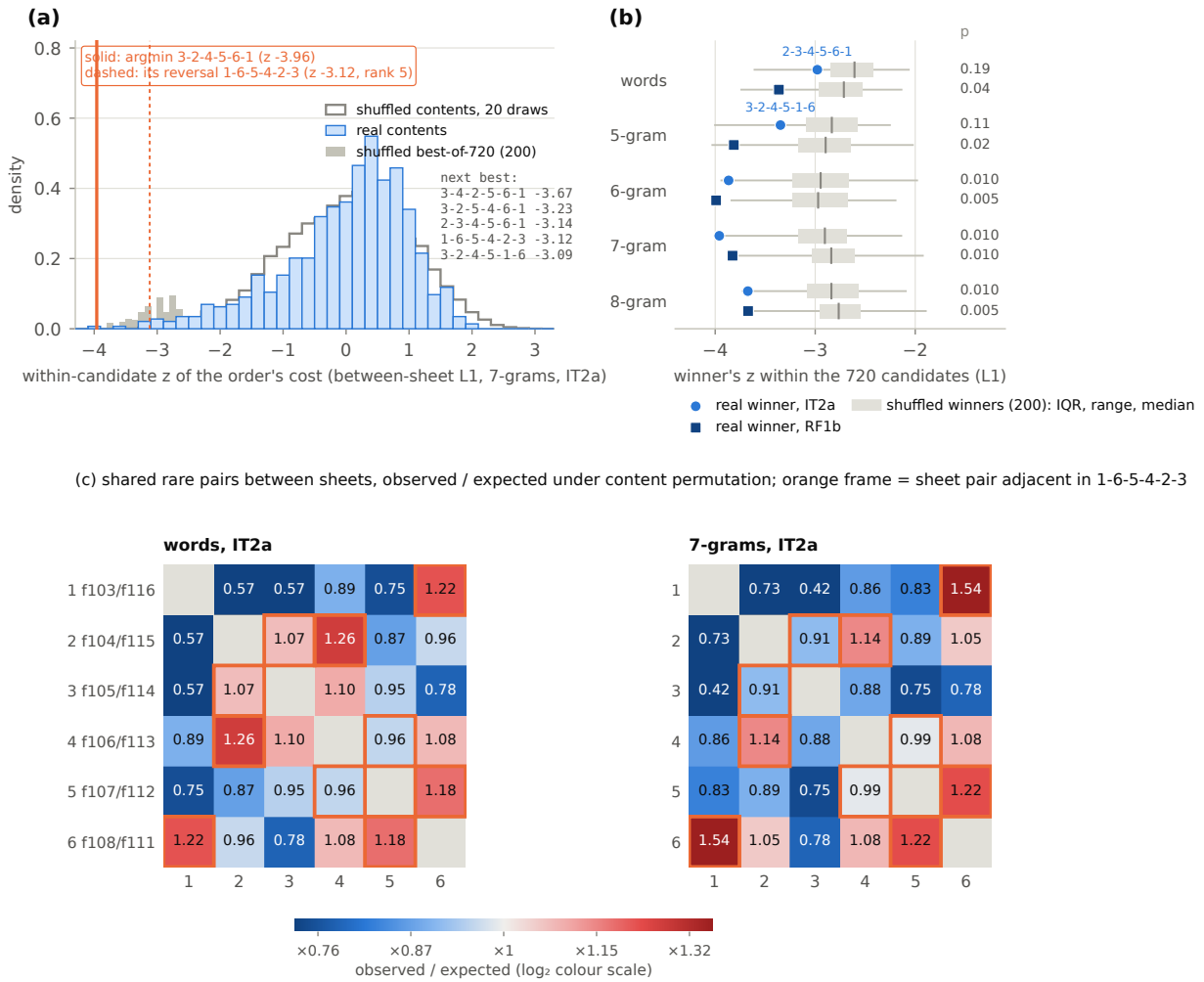


Figure 12. Quire T. (a) Histogram of the within-candidate z of the 720 stacked orders on the real contents and on shuffled contents, with the winner 1-6-5-4-2-3 and its sheet-reversal marked. (b) The winner's z^* per unit and transcription against the distribution of shuffled-content winners, with the empirical p . (c) Between-sheet affinity matrices (shared rare pairs, observed/expected; words and 7-grams, IT2a) with the chain's adjacencies framed. The recorded JSON holds summaries only, so the candidate vectors and affinities were re-derived with the recorded code (figures/derive_quire_T.py, output data/derived_quire_T_costs.json, fresh null seed); the recorded $z^* = -3.96$ for $n = 7$ under L1 is reproduced exactly. [record: §15, 17]

sheet in that order; or a chain of sheet-level subject affinities that happens to run outermost $\rightarrow$ innermost $\rightarrow$ outward. The test cannot tell the two apart. [record: §15]

7 Other ways of gathering the sheets

Sections 4 and 6 compared the stacked orders with one nested arrangement, the binding. Two alternatives were left open: the same sheets (bifolia) nested in a different order, and a quire made of two or three smaller nested gatherings laid one after another. Both would keep some conjugate leaves apart and might describe the rare-text locality as well as the stacked optimum does. Here every such pattern is a candidate. [record: §19]

Pattern space. An order σ of the S sheets is cut into consecutive blocks, and each block is nested: its sheets' a -leaves outer $\rightarrow$ inner, then their b -leaves inner $\rightarrow$ outer. A block of one sheet is that sheet read stacked, so the fully stacked orders (S blocks) and the fully nested ones (one

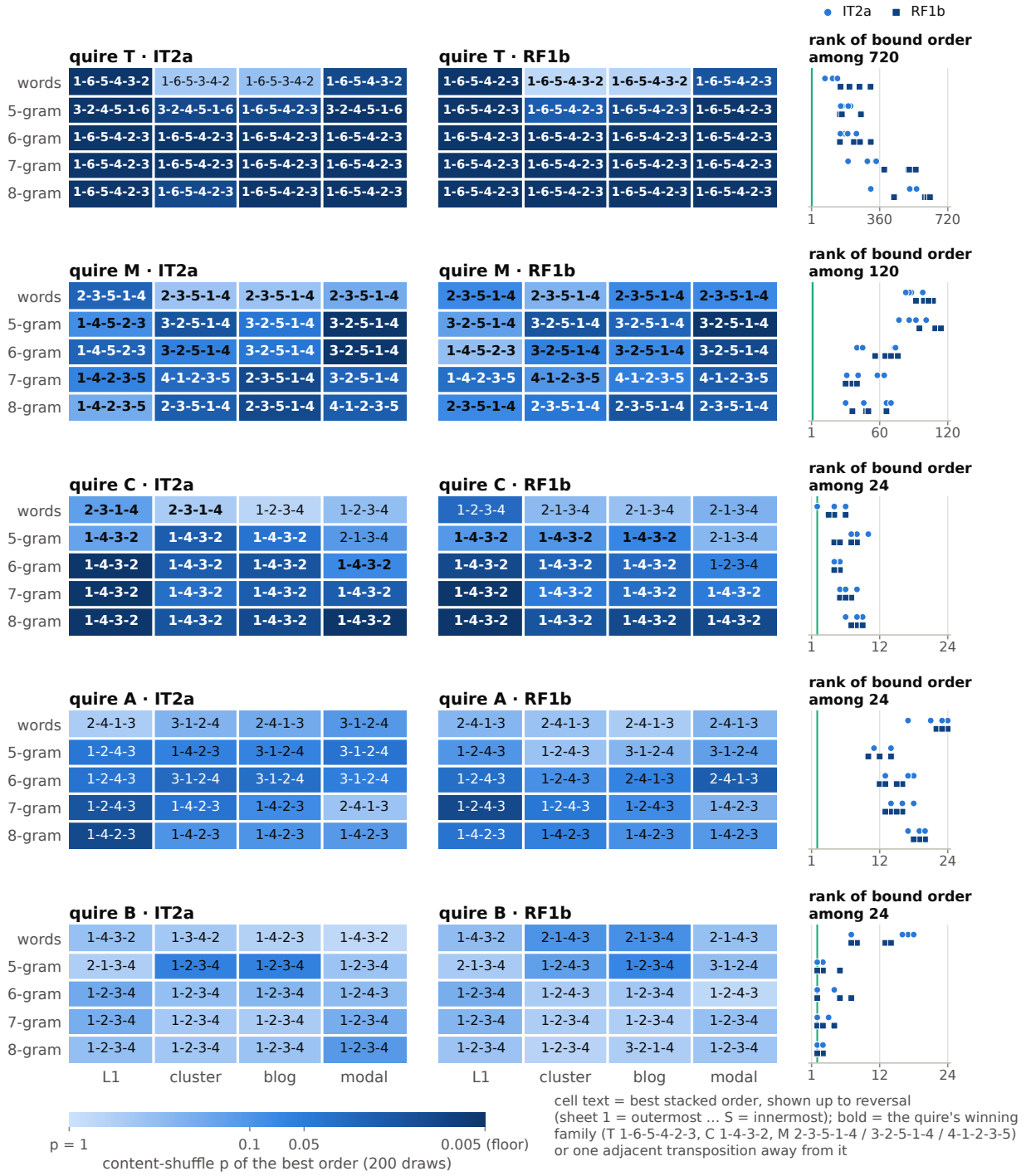


Figure 13. All five quires: content-shuffle p_{best} of the winning order per unit and cost (log colour scale; floor 1/201), both transcriptions, with the winning order and the rank of the current outer→inner order. [record: §14, 15]

block, the binding among them) are the two extremes of one space of

$$N_S = S! 2^{S-1} \quad (11)$$

page sequences: 192 for a four-sheet quire, 1,920 for M, 23,040 for T (quire B has 144 distinct sequences, because its fourth “sheet” is the single leaf f13 and a block that starts with it reads the same as stacking it before the block). Labels write blocks between bars with sheets outer→inner inside a block: 1 | 4 | 3 | 2 is stacked, 1432 nested, 14 | 32 two nested pairs.

Cost, and a bias to declare. A nested block separates a sheet’s two leaves, so the within-sheet distances are no longer constant across candidates and the pair set of Section 6.1 cannot be reused. The cost for this question is the mean distance over *all* cross-page occurrence pairs of rare types,

$$c_{L1}(\pi) = \frac{1}{|P|} \sum_{(i,j) \in P} d_{\pi}(i,j), \quad (12)$$

the “stacked at all versus as bound” setting of Section 6 extended to the whole space. The cluster-scaled costs were also run, with a home-cluster term added (every home occurrence to its nearest other home occurrence) that charges any pattern splitting a home sheet. That family is **structurally biased toward stacked patterns** here: on shuffled contents its winner is fully stacked in 76–100 % of draws on the n -grams, because the home sheet is by definition the sheet holding most of a type’s occurrences and keeping its four pages contiguous minimises the home term whatever the content; and nested-written prose is never ranked first by it in 27 controls. The L1 cost has no such bias—its shuffled-content winners fall on the block classes in proportion to class size (fully nested winners 6–7 % of null draws in M against 120/1,920 = 6.3 % of the space; 12–17 % in the four-sheet quires against 24/192 = 12.5 %) and it recovers nested writing when prose is written nested (below). So L1 is the cost for the class question and the cluster family is used only as a check of the ranking within the stacked class, where it reproduces Section 6.3. The null is the content shuffle of Section 6.2 (200 draws; 100 on T) with the best of all patterns recomputed each time; three statistics are compared with their own null: the winner’s p_{best} , the *gain* of the full space over the best stacked order (in candidate sd), and the *margin* by which the best fully nested order trails the best stacked order, with $P(\text{chance})$ the fraction of shuffles producing a margin at least as large.

Power. Known prose written onto each quire’s page slots (three texts $\times$ three windows, words, L1). Written as bound, the binding ranks 1 of 192 in 2/9 (C), 1/9 (A) and 3/9 (B) cases, 0/9 of 1,920 (M) and 1/9 of 23,040 (T); written nested in a random sheet order, that order ranks 1 in 4/9, 0/9, 2/9, 0/9, 1/9; a random two-block pattern 1/9, 0/9, 1/9, 1/9, 1/9; a random stacked order 1/9, 1/9, 0/9, 1/9, 1/9. Median ranks are 9–27 in every mode (T’s stacked-random 134 of 23,040). The cost identifies the writing *pattern* about as well as it identified the writing *order* in Section 6.2, with no preference among the four kinds: prose written nested is won by a nested pattern, prose written stacked by a stacked one. [record: §19]

Results. Table 8 and Figures 14 and 15. **Quire T (23,040 patterns) is the decisive case.** A fully stacked order wins the whole space on every n -gram unit and both transcriptions—3 | 2 | 4 | 5 | 6 | 1, the un-folded optimum of Section 8, on seven of the eight cells and 3 | 2 | 4 | 5 | 1 | 6 on the eighth (IT2a, $n = 5$)—at $p_{\text{best}} = 0.01$ against the shuffled winner. On the n -grams the best fully nested order ranks 5,007–10,535 (IT2a) and 6,146–8,434 (RF1b), the binding 19,180–22,405 and 17,902–21,492, and the nested class trails the stacked optimum by 3.4–4.7 sd where shuffled contents give +0.7 to +0.9 ($P(\text{chance}) = 0.01$ on all eight n -gram cells, 0.02–0.05 on words). Nothing in the full space improves on the best stacked order: the shuffled winner typically beats its own best stacked order by 0.9–1.2 sd through the wider search, the manuscript’s by 0.00 on the n -grams and 0.16–0.18 on words (p 0.90–0.95). In every quire and both transcriptions the overall winner is fully stacked or a pattern with one nested pair among otherwise stacked sheets (A’s 1 | 42 | 3, M’s 14 | 5 | 2 | 3, C’s 2 | 13 | 4 on words), and where it is not fully stacked it beats the best stacked order by 0.02–0.4 sd, exactly what the enlarged search gains on shuffled contents (p 0.7–1.0 in every n -gram cell; the only values below 0.5 are quire B’s words, a null cell where nothing is significant, p_{best} 0.4–0.7). No fully nested order comes close: the best of them ranks 14–106 of 192 in the herbal quires (B’s words excepted), 177–881 of 1,920 in M and

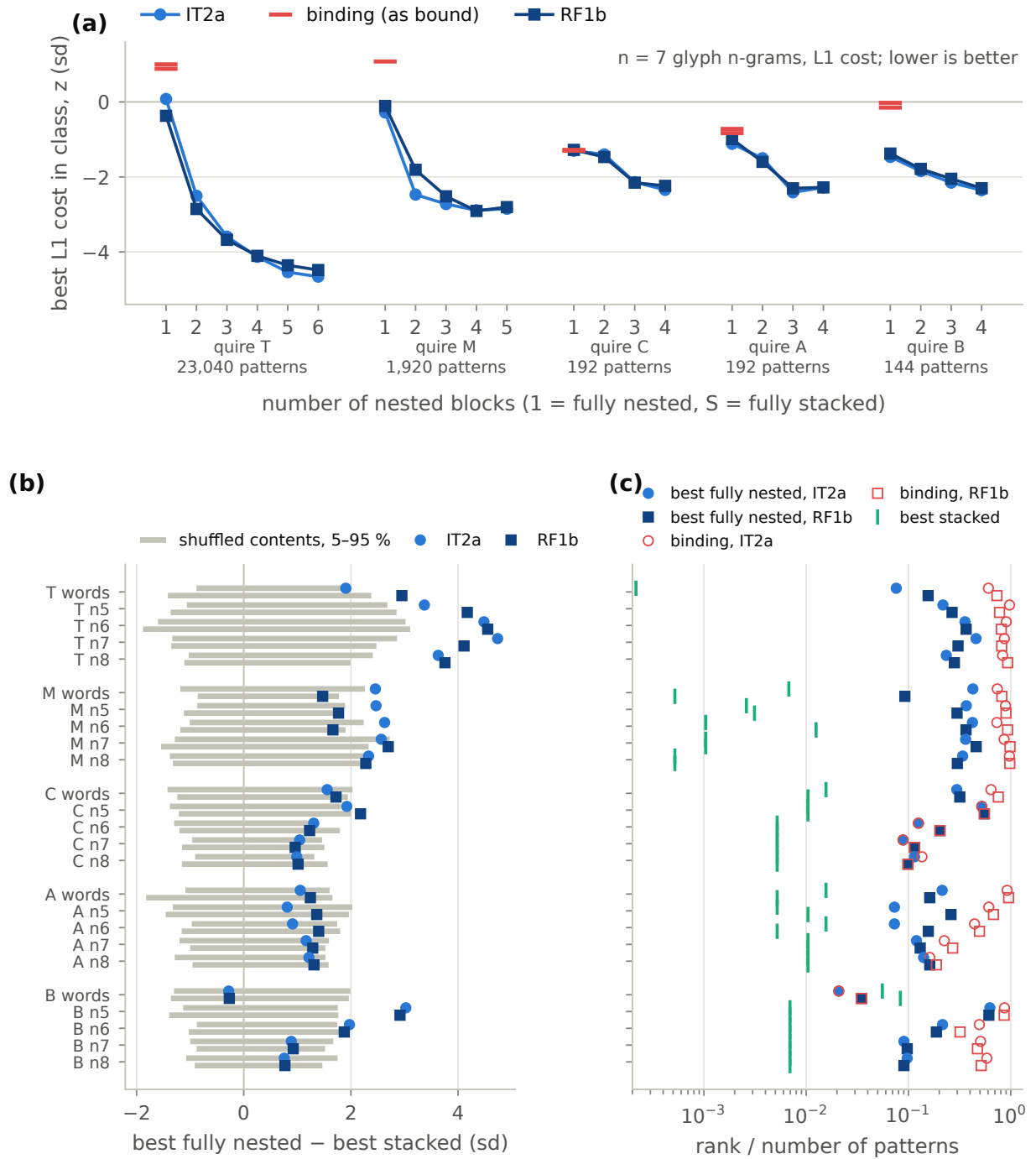


Figure 14. Every nesting pattern of a quire's sheets, L1 cost. (a) Best cost per number of nested blocks (1 = fully nested, S = fully stacked), standardised within each quire's candidate set, $n = 7$ glyph n -grams; the binding's own value is the red tick. The cost falls monotonically toward the stacked end in every quire and on both transcriptions. (b) Margin by which the best fully nested order trails the best stacked order, per quire and unit, against the 5-95 % band of the same margin on shuffled contents. (c) Rank of the best fully nested order (filled) and of the binding (hollow red) among all patterns, with the best stacked order's rank (aqua tick). [record: §19]

Table 8. Every nesting pattern of a quire’s sheets (L1 cost, Equation (12); ranges over the five units words, $n = 5-8$; both transcriptions). “Blocks in winner” is the number of nested blocks in the overall winner per unit ($S =$ fully stacked). “Gain” is how far the overall winner lies below the best stacked order, in candidate sd, with the fraction of shuffled contents whose winner gains at least as much (a gain at $p \geq 0.7$ is the ordinary effect of searching 8–32 times more candidates). “Margin” is how far the best fully nested order trails the best stacked one, with $P(\text{chance})$ from the same shuffles. [record: §19]

quire (S, N_S)	transcr.	blocks in winner words/n5/n6/n7/n8	gain over best stacked, sd (p)	rank of best stacked	rank of best fully nested	rank of binding	margin nested – stacked, sd (P)
A (4, 192)	IT2a	3/4/3/3/3	0.00–0.13 (0.80–1.00)	1–3	14–41	31–177	+0.8 to +1.2 (0.12–0.36)
	RF1b	4/3/4/3/3	0.00–0.08 (0.90–1.00)	1–2	25–50	36–183	+1.2 to +1.4 (0.09–0.15)
B (4, 144)	IT2a	2/4/4/4/4	0.00–0.79 (0.33–1.00)	1–8	3–90	3–125	–0.3 to +3.0 (0.00–0.74)
	RF1b	2/4/4/4/4	0.00–0.69 (0.40–1.00)	1–12	5–88	5–124	–0.3 to +2.9 (0.00–0.74)
C (4, 192)	IT2a	3/3/4/4/4	0.00–0.32 (0.71–1.00)	1–3	17–100	17–123	+1.0 to +1.9 (0.05–0.16)
	RF1b	3/3/4/4/4	0.00–0.02 (0.92–1.00)	1–2	19–106	19–145	+1.0 to +2.2 (0.03–0.23)
M (5, 1,920)	IT2a	4/4/4/4/5	0.00–0.38 (0.71–1.00)	1–13	650–818	1,404–1,862	+2.3 to +2.6 (0.00–0.07)
	RF1b	5/4/3/4/5	0.00–0.41 (0.72–1.00)	1–24	177–881	1,569–1,887	+1.5 to +2.7 (0.03–0.10)
T (6, 23,040)	IT2a	4/6/6/6/6	0.00–0.18 (0.90–1.00)	1–5	1,750–10,535	13,942–22,405	+1.9 to +4.7 (0.01–0.05)
	RF1b	5/6/6/6/6	0.00–0.16 (0.95–1.00)	1–2	3,588–8,434	16,868–21,492	+3.0 to +4.6 (0.01–0.02)

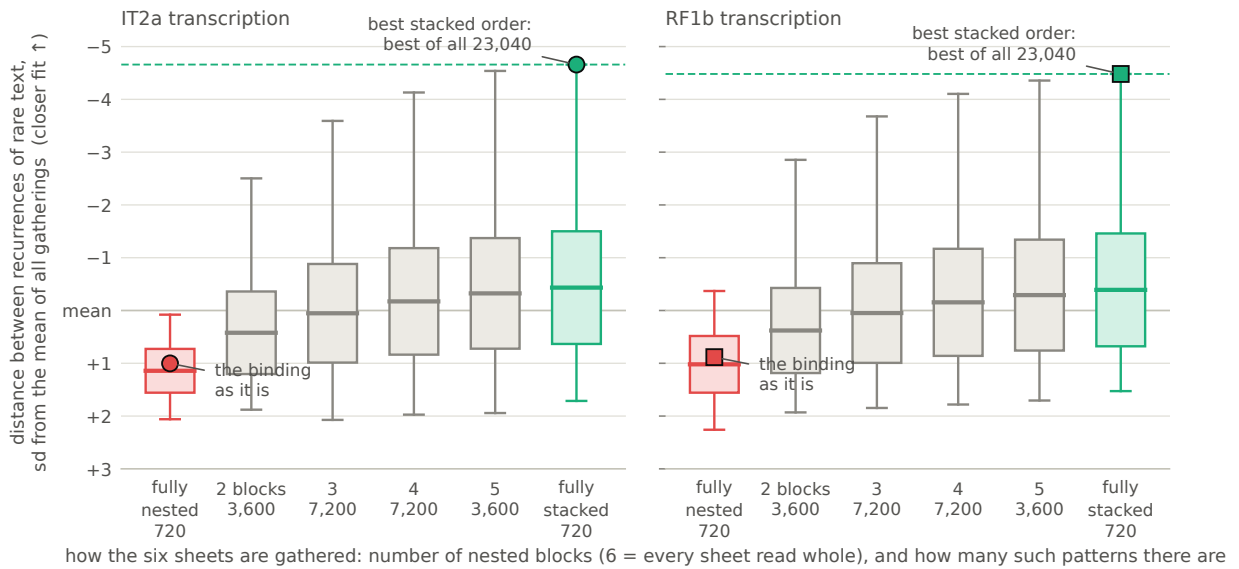


Figure 15. Every nesting pattern of quire T’s six sheets by class, L1 cost on $n = 7$ glyph strings standardised over all 23,040 patterns (closer fit upward), one box per number of nested blocks: mean ± 1 sd, mean line, whiskers the worst and best pattern of the class. The binding and the best stacked order (the best of the whole space; dashed line) are marked. The fully nested class is tight and wholly on the worse side of the mean (its best member +0.08 sd IT2a, –0.37 RF1b); the stacked class is centred better but broad (mean –0.4 sd, sd 1.07), so the classes overlap and a random stacked order beats a random nested one in 95 % (IT2a) and 90 % (RF1b) of pairs; mean and best member improve monotonically with the number of blocks, and the best mixed patterns (4–5 blocks, one or two sheets nested) come within 0.1–0.5 sd of the stacked optimum. [record: §19]

5,007–10,535 of 23,040 in T, and the binding ranks lower still. The nested class trails the stacked optimum by 1–3 sd in the smaller quires, which shuffled contents reproduce in 3–15 % of draws on most n -gram cells (M IT2a 0.005–0.07, C 0.03–0.23, A 0.09–0.36). Two-block patterns—the natural “two smaller gatherings” reading—never win on an n -gram unit anywhere. [record: §19]

Reading. The comparison of Sections 4 and 6 was not a straw man. What the rare text rewards is the property every stacked order shares, a sheet’s four pages together, and no other way of gathering the same sheets—reordered nesting, two or three nested sub-gatherings, or a mixed

Table 9. Direction test: un-folded winner, rank of its reversal, gap Δ Equation (14) and its selection-matched null. [record: §16]

quire	un-folded best ($n = 5-8$, both transcriptions; L1 · blog · modal)	reversal's rank	Δ (sd)	null Δ (sd)	p
T	3-2-4-5-6-1 in 22 of 24 L1·blog·modal cells (29 of 32 including cluster); exceptions IT2a $n = 5$ L1/cluster/modal 3-2-4-5-1-6; words 2-3-4-5-6-1 / 3-2-4-5-6-1	5-27 / 720	0.6-1.7	0.8-1.4 $\pm$ 0.5-0.7	0.20-0.80
M	inconsistent: 2-3-5-1-4, 4-1-5-2-3, 1-4-5-2-3, 1-4-2-3-5, 4-1-2-3-5	2-35 / 120	0.1-2.1	0.8-1.5	0.07-0.98
C	1-4-3-2 ($n = 6-8$), 2-3-4-1 ($n = 5$)	2-6 / 24	0.05-1.1	1.1-1.6	0.66-1.00
A	3-1-2-4 / 1-2-4-3 / 1-4-2-3, split	2-8 / 24	0.0-0.9	1.0-1.5	0.55-1.00
B	1-2-3-4 and 4-3-2-1 both appear	2-9 / 24	0.0-1.5	1.0-1.3	0.38-1.00

pattern—recovers it. Within the stacked class the T chain of Section 6.3 stands as recorded (adjacency only; direction in Section 8). Two limits. The cluster-family costs cannot be used for the class question at all (the bias above), so this section rests on L1 alone. And the space does not contain inverted folding combined with nesting, which nobody has proposed; a nested pattern whose leaves coincide with the stacked adjacency is already in it, since a sheet nested inside a one-sheet block is that sheet stacked. [record: §19]

8 Direction

8.1 An order and its sheet-reversal are not equivalent

When the seriation results were first recorded, an order and its sheet-reversal were treated as equivalent and each best order reported as the smaller label of the pair. That is wrong: reversing the order of the sheets (bifolia) does not reverse the page sequence, because every sheet is still read a^r, a^v, b^r, b^v . For a type with an occurrence at within-sheet position x in sheet A (length L_A) and one at y in sheet B (length L_B),

$$d_{AB} = (L_A - x) + y, \quad d_{BA} = (L_B - y) + x, \quad d_{AB} - d_{BA} = (L_A - L_B) - 2(x - y), \quad (13)$$

so shared types falling *late* in one sheet and *early* in the next favour that reading sense. Summed over pairs, this asymmetry is the direction signal; it is a small part of the between-sheet distance (the within-sheet position, against whole-sheet offsets of ≈ 700 tokens), and the adjacency results of Section 6 are unaffected (reversal pairs differ by ≤ 2 sd and were always ranked together).

[record: §16]

8.2 The gap to the reversal and its null

For each unit and cost the un-folded winner σ^* , its reversal $\bar{\sigma}^*$, the reversal's rank, and the gap

$$\Delta = \frac{c(\bar{\sigma}^*) - c(\sigma^*)}{s_c} \quad (\text{in candidate sd}) \quad (14)$$

are computed. Because *any* winner beats its reversal by construction, the null for Δ is taken from each content shuffle's own winner and *its* reversal (200 draws)—the same selection, so the selection bias is inside the null—and p is the fraction of shuffles with a gap at least as large.

[record: §16]

No quire’s direction is established (Table 9, Figure 16a): every gap between the best order and its reversal is the size that selecting the best of the candidate set produces on shuffled contents ($p \geq 0.07$, mostly ≥ 0.3). Quire T’s direction is at least *consistent*—3–2–4–5–6–1 in 22 of 24 L1·blog·modal cells (29 of 32 with cluster; the exceptions are the IT2a $n = 5$ cells, 3–2–4–5–1–6), i.e. f105/f114, f104/f115, f106/f113, f107/f112, f108/f111 and the outermost f103/f116 *last*—but the cells are correlated and the gap is ordinary; and “the outermost sheet holds the end” is also where a text written as bound ends (f116). [record: §16]

8.3 Is there a time arrow inside a cluster?

The only route to direction that does not depend on the sliver of within-sheet position in Equation (13) would be cluster *shape*: if a rare type is introduced, reused quickly and tails off, a cluster straddling a sheet boundary tells from its shape which sheet came first. For every type with 3–10 occurrences inside one segment, two statistics were taken: the fraction of types whose first inter-occurrence gap is shorter than the last (front-loaded $\Rightarrow > 0.5$; sign test) and the skew $(\bar{x} - \frac{x_1+x_k}{2})/(x_k - x_1)$ (front-loaded $\Rightarrow < 0$; t -test). Segments were 1,500-token windows of six known texts (words and character 5–8-grams), the manuscript’s 50 sheets of ≥ 3 pages read a^r, a^v, b^r, b^v (direction known, no order search), by Currier language, and its pages; the control puts the four pages of every four-page sheet in random order (200 draws), keeping every page-level effect and destroying only the reading order. [record: §18]

Prose has no usable arrow. Pooled over six texts, the first<last fraction is 0.503 for words ($n = 19,551$) and 0.502–0.508 for n -grams, skew -0.0004 to -0.0011 ; by text the fraction ranges 0.48–0.53 with both signs. The only trace is in long n -grams with ≥ 5 occurrences (0.52–0.55), a few per cent. A cluster is, to within a per cent, symmetric in time, so the premise fails and the shape of a straddling cluster cannot give direction anywhere. The manuscript is symmetric too (all sheets pooled 0.49–0.51; pages 0.48–0.51; Currier *B* sheets 0.49–0.50, permutation p 0.17–0.61). The one asymmetry—Currier *A* herbal sheets, n -grams only, back-loaded at 0.44–0.47, driven by $k = 3$ types, p 0.005–0.1 against the within-sheet page permutation on six correlated cells, both transcriptions—is not the mirror of any prose property (prose $k = 3$ is 0.50 exactly) and so is not evidence that *A* sheets are read backwards; its shape is what a *leaf-level* difference would produce (the two pages of leaf *b* sharing rare n -grams more than the two pages of leaf *a*), a layout or production question given that herbal a^r pages carry the large drawings. Page length in *A* sheets falls $104 \rightarrow 95 \rightarrow 87 \rightarrow 82$ tokens by position, but the permutation null carries lengths with the pages and still gives 0.50. [record: §18]

9 Shape of the nulls

The “ $\pm sd$ ” summaries and the best-of- N heuristic were checked against the actual null distributions on quires T, M and C (IT2a, 200 content shuffles, L1 and modal). [record: §17]

- **Candidate cost vector.** On shuffled contents it is near-normal but mildly flat: skew ≈ 0 , excess kurtosis -0.3 to -0.8 ; Shapiro–Wilk rejects at 5 % in 6–34 % of shuffles for M (120 orders) and C (24), and in 80–92 % for T (720 orders, where the test has power against the flatness). On the *real* contents T’s vector is clearly non-normal (skew -0.4 to -1.0 , kurtosis up to $+0.8$ at $n = 6$ –7): a left tail of low-cost orders—the winning family itself—which is the signal, not a null defect. M’s real vectors are mildly left-skewed (-0.1 to -0.6), C’s mildly right-skewed.
- **Winner’s within-candidate z .** Median z^* of the shuffled winner: M -2.26 to -2.42 (Gaussian heuristic -2.40), C -1.80 to -1.91 (-1.75), T -2.65 to -2.98 (-2.99). The heuristic was adequate. Empirical p of the real winner’s z^* against the shuffled winners’: T $n = 6$ –8 p 0.005–0.015 (both costs), $n = 5$ 0.045 / 0.16, words 0.13–0.15; M modal $n = 5$ 0.005, $n = 6$

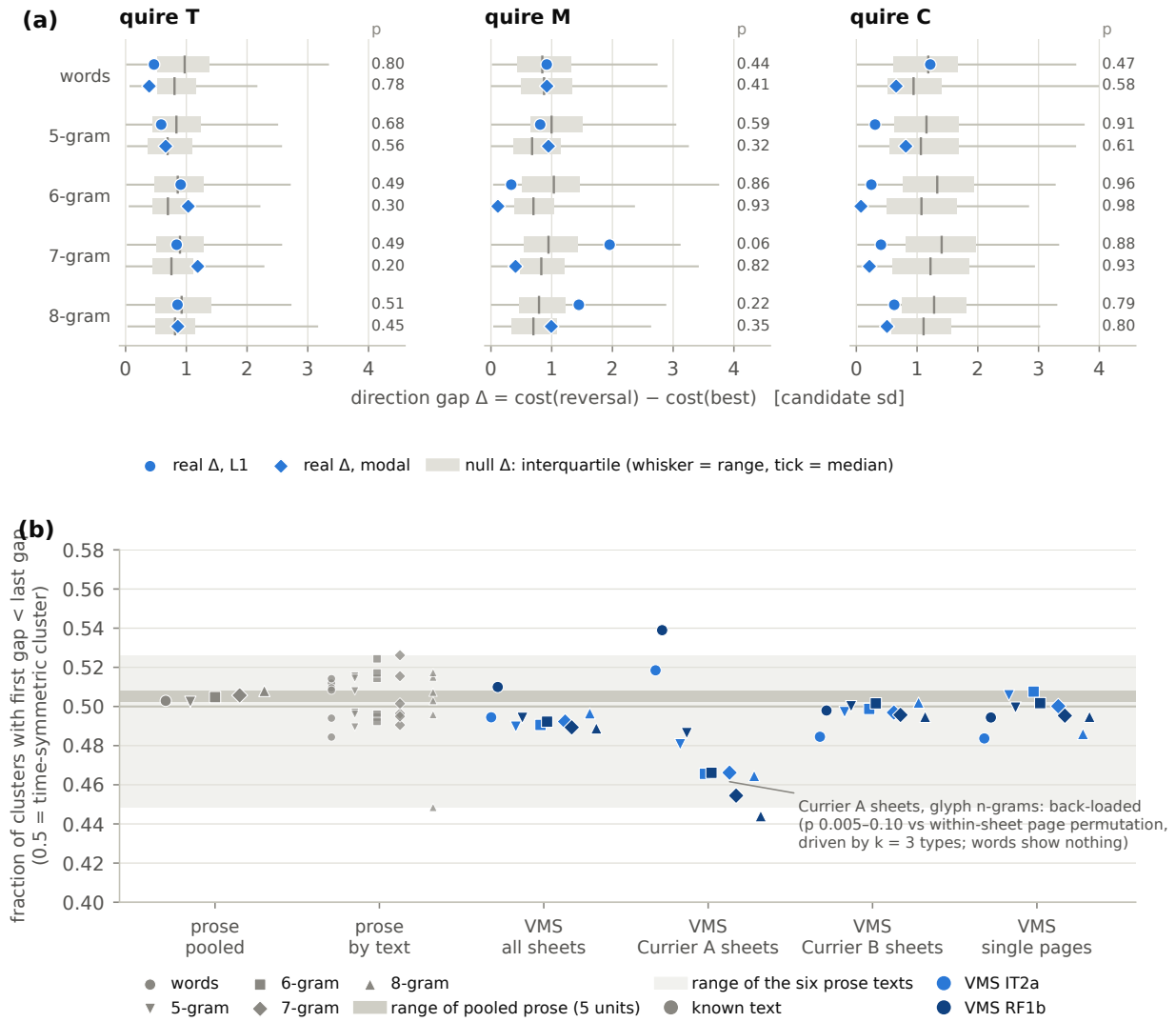


Figure 16. Direction. (a) Gap Δ between the winning order and its sheet-reversal (in candidate sd) against the selection-matched null from shuffled contents, per quire and unit. (b) Cluster front-loading: fraction of rare types whose first gap is shorter than the last, for known texts and for the manuscript's sheets and pages; 0.5 is symmetry. The two grey bands are the range of the pooled prose value over the five units and the range of the six individual prose texts. [record: §16–18]

0.07, the rest 0.16–0.51; C 0.50–0.97—matching the p_{best} values of Table 7, which were already empirical.

- **Direction gap Δ .** Bounded at zero and right-skewed (skew +0.2 to +1.3; median 0.7–1.4 sd, upper quartile 1.0–2.0, maxima 2.2–4.0), so “mean $\pm$ sd” was a poor summary; the empirical p of Table 9 stands. The real Δ falls inside the null interquartile range on every T cell (p 0.20–0.83) and on C (p 0.48–0.99); M's only low cell is $n = 7$ L1 at p 0.085.

Nothing in Sections 6 and 8 rests on normality: the candidate distribution's departure from normal in T is the finding, and the gap distribution's departure is why only the empirical p was ever quoted.

10 Discussion

What this establishes

1. The manuscript's rare text has about one tenth of the passage-scale locality of prose and clusters like rhymed verse at every frequency class $k = 2-5$ and in both Currier languages (Section 3).
2. Despite that, the sheet (bifolium) is a unit that shares rare text: conjugate leaves share more, nested neighbours share less, on words and on glyph n -grams, on both transcriptions, with quire, language and hand held fixed (Section 4). Six real texts reproduce the sign and size only when written sheet by sheet (36/36 vs 36/36). No other way of gathering the same sheets does as well: over every reordered, sub-gathered and mixed nesting of a quire's sheets (up to 23,040 patterns) nothing beats the best stacked order beyond what the wider search gains on shuffled contents, the best fully nested order ranks 5,007–10,535 of 23,040 in T, and the binding lower still (Section 7).
3. The effect is small in mass—rare text is not contained by sheet, solver windows are unchanged, no sheet order lifts the locality above the noise ceiling (Section 5)—and lives in the $\sim 20\%$ of pairs that fall within a quire.
4. Within a quire the cluster-scaled seriation selects a sheet chain in T (1–6–5–4–2–3, 31/32 cells, $p \leq 0.02$ against shuffled contents and geometry), replicates the shape in C (1–4–3–2) and is consistent in M; in each the outermost sheet neighbours the innermost—informative for T (chance rate $2/S = 1/3$; nested-written prose $1/9$), at chance ($1/2$) for the four-sheet quires on its own, so C's contribution is the full chain rather than this feature (Section 6).

What it does not establish

1. **Composition unit versus work unit.** Every statistic here responds equally to “the text flows $a^r \rightarrow a^v \rightarrow b^r \rightarrow b^v$ across the sheet” and to “one scribe, one session, one topic per sheet, with the intended reading order still nested”. The hands running by sheet (97% of conjugate pairs same hand) is itself evidence that the sheet was a working unit. A rare-text statistic cannot separate the two; that is what Davis's stacked proposal and a sheet-level production account share.
2. **Direction.** The sense of reading along any chain is not established (Section 8): the gap to the reversal is what selection produces on noise, and prose clusters are symmetric to within a per cent, so there is no arrow to calibrate on.
3. **Which sheets or quires are misplaced.** The per-quire word-level picture is uneven (signal in A, T, H, M; none in herbal B–G); quires D–G (mixed language/hand), O, Q and S were not seriated; the codicological literature's specific reorderings were not tested as fixed candidates; T's chain has not been checked against the codicology (f103/f116 is the sheet with the missing f116v text).
4. **A stacking that recovers prose-like locality.** None exists: the separation from prose is within the sheet (Section 5.3).

Sequential testing and correlated cells. The evidential order matters. The doubleton test was null; the pooled-word statistic was chosen after seeing it ($p \approx 0.01$, not pre-registered); the n -gram test was then run as the pre-stated check and is the number to quote ($z = +3$ to $+5$, 0/5,000 even-spread draws at $n \geq 7$, hand-controlled). Within every table the cells are correlated: n -grams of neighbouring n overlap, the two transcriptions are one text, the three cluster-scaled costs share their outliers, the nesting-pattern cells of Section 7 share their candidates with the stacked seriation, and the herbal quires' candidate sets are tiny (24 orders), so their p -values are

coarse. Quire T’s seriation is the result to quote; C is a replication in a different language, hand and section; M is consistent; A and B are uninformative.

What would settle it. (i) An independent quire-internal signal reaching the same optimum—the words alone at $p < 0.05$ in T or M, or a whole-vocabulary leaf similarity with language and section controlled. (ii) The specific reorderings proposed in the codicological literature run as fixed candidates against the within-quire null, instead of a search over all $S!$. (iii) The remaining multi-sheet quires. (iv) A production-side test of the composition-vs-work distinction that does not use text position at all (ink, ruling, drawing sequence). (v) For the Currier-A n -gram back-loading, the one follow-up worth running before any interpretation: within-leaf sharing, leaf a versus leaf b , on the herbal quires. (vi) The latent-semantic-analysis test Davis reported (Section 1), should it be published: a page-level topic similarity is the natural complement to a rare-text position statistic, and agreement or disagreement on the sign of the conjugate-versus-nested contrast would be informative either way. None of this needs the GPU; every run here is seconds to minutes on CPU.

A Reproduction

All artifacts referenced below are copied into the report folder’s `data/` directory (from the analysis root’s `analysis/doubleton_gaps/`); the analysis scripts live in the repository’s `scripts/` directory and are run as

```
uv run python scripts/<name>.py
```

Figures are rebuilt with `make figures` (one script per figure under `figures/`, shared style in `style.py`); the document with `make`.

Table 10. Analysis scripts, their artifacts and the sections they support.

script	artifacts (data/)	report section (record §)
<code>doubleton_gaps.py</code>	<code>summary.json</code> ; <code>vms*_doubletons.csv</code> ; <code>page_affinity*.csv</code>	Section 3 (2, 3, 4)
<code>doubleton_leaf_affinity.py</code>	<code>leaf_affinity.json</code> ; <code>leaf_affinity_controls.json</code>	Section 4 (4)
<code>rare_type_clustering.py</code>	<code>rare_types.json</code> ; <code>rare_types_controls.json</code> ; logs	Sections 3 and 4 (6)
<code>glyph_ngram_leaf_test.py</code>	<code>glyph_ngrams.json</code> ; <code>hand_control.json</code> ; logs	Sections 4 and 4.4 (7)
—	<code>window_tokens_per_type.log</code> ; <code>containment.log</code> ; <code>corpus_sweep.json</code>	Sections 3 and 5 (8, 9, 11)
<code>order_optimize.py</code>	<code>order_optimize_none.json</code> ; <code>order_optimize_inverted.json</code> ; logs	Section 5.3 (10)
<code>leaf_test_pvalue.py</code>	<code>leaf_test_pvalue.json</code> ; log	Section 4.5 (12)
<code>quire_order_poc.py</code>	<code>quire_order_poc_M.json</code> ; log	Section 6 (13)
<code>quire_order_burst.py</code>	<code>quire_order_burst_Q.json</code> , $Q \in \{M, T, A, B, C\}$; logs; <code>burst_v1/</code> (archived first runs)	Section 6 (14, 15)
<code>quire_order_direction.py</code>	<code>quire_order_direction_Q.json</code> ; logs	Section 8 (16)
<code>quire_order_nullshape.py</code>	<code>quire_order_nullshape_Q.json</code> , $Q \in \{T, M, C\}$; logs	Section 9 (17)
<code>quire_order_nesting.py</code> (+ <code>_summary.py</code> , <code>_table.py</code>)	<code>quire_order_nesting_Q.json</code> , $Q \in \{A, B, C, M, T\}$; logs	Section 7 (19)
<code>burst_frontloading.py</code>	<code>burst_frontloading.json</code> ; logs	Section 8.3 (18)
<code>burst_frontloading_control.py</code>		

Conventions. Reference null 1 (uniform pairs) is exact for $k = 2$ and Monte-Carlo (200,000 draws) for $k \geq 3$. Page-order shuffles: 1,000 for the page-order statistics (200 per known-text window), 2,000 for the leaf test, 500 for the n -gram controls (1,000 for the hand control and the pooled-word controls), 5,000 for the tail study, 200 for every seriation, direction and null-shape run. The optimiser of Section 5.3 is simulated annealing, 20,000 steps; a first run that allowed a physically impossible leaf swap gave the same numbers and is superseded by `order_optimize_none.json`. The first quire-M proof-of-concept log with a wrong null mask (`quire_order_poc_M_maskbug.log`) is superseded by `quire_order_poc_M.log`.

B Supplementary tables

Table 11. Leaf-pair test on doubletons alone (Section 4), counts by category against the within-quire- and-language null (2,000 shuffles). [record: §4]

category	IT2a obs / null	z	RF1b obs / null	z
conjugate leaves	18 / 14.3 ± 3.6	+1.0	18 / 16.4 ± 3.8	+0.4
nested-adjacent leaves	27 / 27.6 ± 4.7	−0.1	32 / 29.7 ± 4.9	+0.5
innermost sheet (both)	5 / 6.8	−0.7	10 / 9.0	+0.4
other same-quire leaf pairs	101 / 112.2 ± 5.8	−1.9	112 / 125.3 ± 6.1	−2.2
$D = \text{conjugate} - \text{nested-adjacent}$	−9 / -13.3 ± 6.3	+0.7	−14 / -13.4 ± 6.6	−0.1

Table 12. Pooled leaf test, components and per-language split (Section 4): conjugate z / nested-adjacent z / D z . [record: §6.2]

pairs pooled	IT2a	RF1b
$k = 2$ (896 / 997 pairs)	+1.0 / −0.1 / +0.7	+0.4 / +0.5 / −0.1
$k \in [2, 5]$ (4,839 / 5,306)	+2.6 / −1.4 / +2.5	+2.0 / −0.7 / +1.8
$k \in [2, 10]$ (13,665 / 14,625)	+2.2 / −1.8 / +2.4	+2.3 / −1.8 / +2.5
$k \in [2, 10]$, Currier <i>A</i> pages only	+0.7 / +0.2 / +0.3	+1.4 / +0.2 / +0.7
$k \in [2, 10]$, Currier <i>B</i> pages only	+1.1 / −1.4 / +1.8	+1.0 / −1.7 / +1.9

Table 13. Word-level power controls (six texts on the IT2a page slots, 1,000 shuffles): D z , $k \in [2, 5]$ / $k \in [2, 10]$. [record: §6.2]

control	written nested	written stacked
Isidore	−7.1 / −7.3	+3.1 / +3.5
Seneca <i>NQ</i>	−4.2 / −5.5	+1.3 / +2.4
Bullinger	−5.2 / −7.2	+3.3 / +2.8
Decameron	−4.4 / −5.4	+3.6 / +4.7
Commedia	−0.8 / −2.0	+2.0 / +1.5
Orlando Furioso	−3.1 / −3.3	+0.9 / +1.7

Table 14. n -gram power controls (six texts on the [IT2a](#) page slots by symbol count, $k \in [2, 10]$): Dz at $n = 5 / 6 / 7$, all n -grams (boundary-straddling in brackets). Thirty-six of thirty-six nested-written cells negative; thirty-six of thirty-six stacked-written cells positive. [record: §7.2]

control	written nested	written stacked
Isidore	-6.0 / -5.6 / -5.2 (-4.4 / -4.3 / -3.9)	+2.3 / +2.5 / +2.6 (+2.1 / +2.4 / +2.4)
Seneca <i>NQ</i>	-5.5 / -6.2 / -6.4 (-4.9 / -6.5 / -6.0)	+2.4 / +2.9 / +2.4 (+2.0 / +2.6 / +2.3)
Bullinger	-6.3 / -7.6 / -7.8 (-7.0 / -7.2 / -7.3)	+3.2 / +3.4 / +3.7 (+2.3 / +2.6 / +2.9)
Decameron	-5.3 / -5.7 / -5.4 (same)	+4.7 / +4.5 / +4.5 (+4.7 / +4.9 / +4.3)
Commedia	-3.3 / -3.9 / -3.7 (-2.3 / -2.4 / -2.4)	+3.7 / +2.7 / +2.4 (+2.8 / +1.6 / +1.4)
Orlando Furioso	-2.2 / -3.4 / -4.0 (-1.7 / -2.8 / -3.1)	+1.1 / +1.7 / +1.9 (+1.3 / +1.6 / +1.8)

Table 15. Tokens per type in non-overlapping windows, nested $\rightarrow$ stacked reading ([IT2a](#) / [RF1b](#)), with hapax fraction. [record: §8]

stream, W	IT2a	RF1b	hapax fraction
$A, 512$	1.607 $\rightarrow$ 1.620 (+3.2 sd)	1.541 $\rightarrow$ 1.544	0.78–0.80
$A, 1,024$	1.867 $\rightarrow$ 1.875	1.767 $\rightarrow$ 1.769	0.75–0.77
$A, 2,048$	2.192 $\rightarrow$ 2.206	2.035 $\rightarrow$ 2.046	0.73–0.76
$A, 4,096$	2.638 $\rightarrow$ 2.635	2.408 $\rightarrow$ 2.407	0.72–0.74
$B, 512$	1.721 $\rightarrow$ 1.724	1.610 $\rightarrow$ 1.613	0.75–0.78
$B, 1,024$	2.028 $\rightarrow$ 2.032	1.870 $\rightarrow$ 1.882 (+3.2 sd)	0.73–0.75
$B, 2,048$	2.422 $\rightarrow$ 2.430	2.219 $\rightarrow$ 2.205	0.71–0.73
$B, 4,096$	2.972 $\rightarrow$ 2.911	2.661 $\rightarrow$ 2.606	0.70–0.72

Table 16. Quire M, plain-mean (L1) seriation before the cluster-scaled costs (Section 6): best order (reported at the time up to reversal), z^* , p_{best} against shuffled contents, and rank of the current order 1–2–3–4–5, [IT2a](#) / [RF1b](#). [record: §13]

unit	best order	z^*	p_{best}	current order rank / 120
words	2–3–5–1–4 / 2–3–5–1–4	-2.36 / -2.33	0.025 / 0.070	98 / 92
$n = 5$	1–4–5–2–3 / 3–2–5–1–4	-2.55 / -1.94	0.050 / 0.119	86 / 95
$n = 6$	1–4–5–2–3 / 1–4–5–2–3	-2.52 / -1.75	0.035 / 0.383	40 / 65
$n = 7$	1–4–2–3–5 / 1–4–2–3–5	-2.53 / -2.41	0.060 / 0.060	31 / 33
$n = 8$	1–4–2–3–5 / 2–3–5–1–4	-2.34 / -2.42	0.184 / 0.065	46 / 48

Table 17. Quire M, cluster-scaled seriation: best order and p_{best} (content shuffle, 200 draws) per unit and cost, by transcription. The L1 column repeats the plain-mean run of Table 16; where the record lists two orders for one cell they are the [IT2a](#) and [RF1b](#) winners respectively. Bold: $p \leq 0.015$.
[record: §13, 14]

unit	L1	cluster	blog	modal
IT2a				
words	2-3-5-1-4, 0.025	2-3-5-1-4, 0.40	2-3-5-1-4, 0.38	2-3-5-1-4, 0.19
$n = 5$	1-4-5-2-3, 0.050	3-2-5-1-4, 0.015	3-2-5-1-4, 0.045	3-2-5-1-4, 0.005
$n = 6$	1-4-5-2-3, 0.035	3-2-5-1-4, 0.060	3-2-5-1-4, 0.045	3-2-5-1-4, 0.005
$n = 7$	1-4-2-3-5, 0.060	4-1-2-3-5, 0.030	2-3-5-1-4, 0.010	3-2-5-1-4, 0.025
$n = 8$	1-4-2-3-5, 0.184	2-3-5-1-4, 0.020	2-3-5-1-4, 0.010	4-1-2-3-5, 0.020
RF1b				
words	2-3-5-1-4, 0.070	2-3-5-1-4, 0.26	2-3-5-1-4, 0.11	2-3-5-1-4, 0.09
$n = 5$	3-2-5-1-4, 0.119	3-2-5-1-4, 0.005	3-2-5-1-4, 0.020	3-2-5-1-4, 0.005
$n = 6$	1-4-5-2-3, 0.383	3-2-5-1-4, 0.095	3-2-5-1-4, 0.12	3-2-5-1-4, 0.005
$n = 7$	1-4-2-3-5, 0.060	4-1-2-3-5, 0.13	4-1-2-3-5, 0.08	4-1-2-3-5, 0.015
$n = 8$	2-3-5-1-4, 0.065	2-3-5-1-4, 0.045	2-3-5-1-4, 0.030	2-3-5-1-4, 0.035